

Beetles through space and time: Population decline and environmental change over a 20-year timespan in the Cederberg

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KCKCOS001

Dissertation presented for the degree of Master of Science in the Department
of Biological Sciences, University of Cape Town

April 2026

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Abstract

Nearly half of the insect species in the world are projected to become extinct over the next few decades and is thought to occur in most environments globally. Climate change is one of the major driving factors behind insect decline. With the progress of global climate change, the direct effects of increasing temperatures, and the indirect effects of warming temperatures on factors such as soil and vegetation characteristics, are all potential drivers that could explain any changes in insect populations in their respective environments. There is evidence that changing environmental factors can cause suitable habitats for different fauna and flora to shift across landscapes and result in shifting of both the placement and extent of their distribution ranges. This can occur due to novel competition, exceeding critical physiological limits, and availability of food and habitats. Global insect decline is a pressing concern because insects are an ecologically diverse group of organisms that provide a wide variety of crucial ecosystem services to the environments they inhabit. They are also a useful indicator of the status of the other fauna that inhabit the same environments as them and can indicate that those respective environments might experience trophic or ecological collapse. The extent and potential environmental drivers of insect decline are especially understudied in the Southern Hemisphere. This relative lack of research when compared to other taxa has meant that, despite their ecological importance, insects are currently not widely being studied or tracked for conservation strategies in key environments. Therefore, the extent of insect decline in South Africa has not been well-documented. Beetles are the most species rich family of insects globally and exhibit great levels of ecological diversity. They provide crucial ecosystem services such as seed dispersal, decomposition, pest control, acting as prey for other species, and vegetation cover management. The Cape Floristic Region (CFR) is one of the most diverse floral kingdoms globally. The CFR is expected to contract in its extent as climate change progresses, meaning that the insects that rely on specific vegetation present in those environments will have their habitable ranges also contract in upcoming years, potentially resulting in accelerated decline and the ecological consequences resulting from that decline. A useful tool in allowing for effective observational studies on the potential effects of climate change on populations, are

altitudinal gradients. Mountains especially provide a spatially continuous gradient of rapidly changing environmental conditions across their altitudes, allowing for observations of how populations of certain species assemble themselves throughout those changing environments. Analysing these different distributions can allow for the environmental preferences of different species to be extrapolated and can therefore be used as a proxy for how those species could respond to climate change. In this study, we aimed to assess changes in ground-dwelling beetle assemblages across an altitudinal gradient over a 20-year period. The historical study was performed in 2002, and repeated in October 2022, March 2023, and October 2023 by following the same methodology and locations as previously used. Tenebrionidae, Carabidae, and Cicindelidae beetles were sampled and identified as taxonomically far as possible. For the historical comparisons, three beetle species were compared over time, including *Anthia decemguttata* (Carabidae), *Stenocara dentata* (Tenebrionidae), and *Zophosis gracilicornis* (Tenebrionidae). These are the three most abundant species in this environment and exhibit wide trophic diversity. Along with the beetle sampling, the temperatures in the environment were modelled using meteorological data, soil samples were collected at every site to determine soil characteristics, vegetation ground cover and vegetation height were estimated, and the fire history of the study site was considered. There was a 0.19°C increase in baseline air temperatures, and a 0.84°C increase in ground temperatures over time. Plant litter had increased by 183% and live vegetation had increased by 27% across the 20 years. There was a 97% decline in abundance and an 88% decline in species diversity between the October and March 2023 season. When compared to the historical data, across all three of the focus species there was a 47% decline in abundance overall. Most of this decline in abundance was due to population decline for the *Zophosis gracilicornis*, which reduced by 66% in abundance between the 2002-2003 samples compared to the 2022-2023 samples. These changes were strongly related to a combination of environmental factors, with increasing air temperatures, ground temperatures, and vegetation, all contributing to a reduced abundance of the species in question. These observations suggest that insect decline is occurring to some extent in this environment. The true extent of its intensity as well as the number of species it affects can only be determined by sampling other beetle families and identifying the present

samples further to match with historical data. Describing the specific biology and ecological relationships of the species in question can also allow for a better understanding of the environmental drivers behind the observed changes. These results indicate that more focus needs to be placed on the beetle populations present within this environment so that, if necessary, the relevant conservation interventions can be put in place.

Acknowledgements

I would like to extend my deepest and most heartfelt gratitude towards my supervisors, Charlene Janion-Scheepers, and Abusisiwe Ndaba. Thank you for all the time, effort, care, and assistance that you gave throughout this project. But most importantly, thank you for the vast amounts of guidance and patience that you granted me through the course of the entire journey. I would not have been able to develop and grow into the researcher I see myself as today, if not for the consideration that was provided to me through the highs and the lows.

To Charlene, thank you believing in me and continually reinforcing the importance and relevance of the work that I have been doing. I would not have been able to push myself through all the hurdles that I encountered if it were not for your motivational direction. I am extremely grateful for all the abundant, but small ways in which you have improved me as a researcher and academic. I cannot begin to count the number of times I was struggling with something silly, and you presented me with a simple and elegant solution that solved all my struggles. I have improved my scientific reading, research, editing, and many other skills because of you.

To Abusisiwe, thank you for all the help that you provided me in the lab over the last few years. You were the one who taught me how to pin, mount and label all my samples. None of the 1,633 beetles that I sorted would be so, if you did not give me the rigorous training that you did. Thank you also for all the additional sorting and processing that you did for the further samples that were collected. You were always available to help me check my data analyses and work through our modelling together. Only thanks to working together with you, all the necessary information for this project was captured, and all the gaps in those data were closed.

Thank you to the Centre for Invasion Biology, and to Suzaan Kritzinger-Klopper, Erika Nortjé, Calvin Hartnick, and all the volunteers, who collectively planned, managed, and carried out all the fieldwork. You all collected the samples and data that allowed this amazing study to be conducted in the first place. Your labour has allowed for this greatly valuable dataset to be created. Regardless of my slice of this whole study, the Cederberg project would not exist at all without all of you.

To Dr Werner Strümpher, thank you for accepting and carrying out the task of identifying the species in my samples. Where my taxonomic knowledge was failing me, your assistance allowed for amazing fast progress.

I would like to thank the Iziko Museums of South Africa (Cape Town, South Africa) for giving me full access to their resources and to work with their beetle collection to allow for the initial identification of my samples. Thank you to Aisha Mayekiso for the warm welcome to the museum and helping me to get started in my searches through the collection. Thank you to Mujahid Hector for the patience and help he showed me during my entire time working at the museum, and for helping me find the samples I was looking for both in the database and in the collection.

Thank you to the entire Soil Ecology & Entomology Research Group at the University of Cape Town for always providing a helpful, friendly, and welcoming environment within our lab. Thank you to Gemma Walker for providing me transport to and from the ICSZ 2024 conference, and for the long talks we had on those rides. Bonding and sharing our experiences of work together helped reassure me through many periods of doubt. Thank you to Juliette Chassain for always being available to talk or hear me out, especially in moments when I was feeling intimidated or uncertain. Your showings of friendship and dismissal of my silly worries helped me more than you may ever know. Thank you to Dean Erasmus for helping me through problems both statistical and mental. The acceptance and understanding you showed as we were both progressing through a similar journey helped me to adjust my own expectations and allowed me to be kinder to myself.

To Cheryl Inggs, thank you for providing a space where I could talk and work through my emotions. Your words of encouragement kept my confidence topped up through all the struggles that I experienced. To all my personal friends, thank you for always being there for me. Whether we were playing together or going out, thank you for listening to me rant, ramble, and otherwise drone on about my work. Thank you to all of you who indulged me by flipping through my dissertation, whether to give detailed feedback or just for reassurance. Thank you to Roffle, and November Kelly. You kept me sane through many days and nights of long, lonely work.

To my partner Sabrina, no words I can ever place upon this page can capture just how much you have helped me through this entire process. You allowed me to keep going and never give up. You provided endless support, emotionally, mentally, practically, and spiritually. From the bottom of my heart, thank you.

Anneme ve babama, siz olmasaydınız bu proje de olmazdı. Maddi, manevi, ruhsal, duygusal, zihinsel, her yönden sizin desteğiniz yüzünden ben bu başarıya sahip oldum. Size sonsuz sevgim var ve bu satırlara ne yazsam yetmez. Babama ve anneme, kalbimin en dibinden teşekkür ediyorum. Şakalarınızdan dualarınıza kadar, hepsi yardımcı oldu.

Ben bu tezi Adnan ve Coşkun dayılarıma anıt olarak yazıyorum. Birisinden isim, öbüründen sevgi elde ettim.

Thank you to the UCT Postgraduate Funding Office for providing me with funding. Thank you to the University of Cape Town for having me as a master's student.

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Chapter 1: General introduction

Insect decline and conservation bias

There has been an estimated 58% decline in global abundance for vertebrates over the last 50 years (WWF, 2016). However, the pace of decline in invertebrates, particularly insects, reported by many studies far exceeds that of vertebrates (Thomas *et al.*, 2004; Hallmann *et al.*, 2017). It is estimated that 40% of the world's insect species could become extinct over the next few decades (Sánchez-Bayo and Wyckhuys, 2019). Studies have already shown that there have been mass declines in insect diversity and abundance in various groups across different countries. For example, a long-term study assessing insect populations in German protected areas reported that flying insect (Lepidoptera and Hymenoptera) abundance declined by 75% in 27 years (Hallmann *et al.*, 2017). Similar results were also observed in the Netherlands, with long-term sampling indicating that moths declined by 61% and ground beetles by 42% across 27 years of monitoring (Hallmann *et al.*, 2020). Outside of Europe, a 36-year study showed declines in biomass, ranging from 78% to 98% depending on the insect group, temperate to tropical regions (Lister and Garcia, 2018). The main proposed reasons for these observed insect declines are due to habitat loss, environmental pollution, pesticides, pathogens, invasive species, and climate change (Sánchez-Bayo and Wyckhuys, 2019).

This proportionately accelerated insect decline has been attributed to various factors. Insects might be more sensitive to gradual small-scale environmental changes that occur in their surrounding environments, meaning their populations respond more quickly to disturbances than those of large organisms (Goulson, 2019). There has also been less effort placed into monitoring and conserving insects compared with other groups of fauna and flora (van der Sluijs, 2020). Public knowledge and perception of the importance of insects have been suggested as a possible reason why there has been such a disparity in conservation efforts (Goulson, 2019). In the most recent United Nations global biodiversity conservation plan, many different vertebrate and plant groups have been selected as conservation targets and indicators for biodiversity health, but insects are still largely overlooked (CBD/COP/15/17, 2023).

Insects provide a wide range of valuable ecosystem services to the environments that they inhabit. Therefore, any declines in diversity or abundance could result in cascading effects across many trophic levels and eventual collapse of the vital ecosystem services that occur in these environments (Wagner, 2020). Most flowering plants are pollinated by animals, and insects make up the majority of these pollinators (Ollerton, Winfree and Tarrant, 2011). In terms of insects' direct effect on humans, 35% of the food produced globally for human consumption relies on insect pollination (van der Sluijs and Vaage, 2016). Almost the entirety of certain essential micronutrients in human diets is provided by these pollinator-mediated crops (Eilers *et al.*, 2011). The decline and partial disappearance of pollinating insects are expected to cause global health problems due to nutritional deficiencies (Daily and Karp, 2015; Smith *et al.*, 2015). In addition to their benefits towards humans, insects also provide vital services to natural environments. They provide natural pest control, decomposition, ecosystem maintenance, and also provide a vast biomass that can feed other organisms present within those environments (Kremen and Chaplin-Kramer, 2007; Oberhauser and Guiney, 2009). Many of these ecosystem functions include highly unique relationships that insects have developed over time with their environments and are not easily replaceable in their respective ecosystems. The decline or disappearance of these services can result in the breakdown of those ecosystems, resulting in mass, multi-level losses in biodiversity (Cardoso *et al.*, 2020).

The prediction of the extent of insect decline, coupled with their environmental importance, implies a necessity to conserve insect taxa that are under threat. However, the baseline data necessary to assess the status of insect decline are not comprehensive globally and are especially lacking in the Southern Hemisphere (Samways, Hamer and Veldtman, 2012). Therefore, studies that track the ecological standing of insects over time could provide insight to assess the risks ecosystems may face regarding insect decline (Chambers *et al.*, 2017).

Compared to the Northern Hemisphere, little is known about the current and predicted effects of climate change on insect biodiversity in the Southern Hemisphere (Chambers *et al.*, 2017). Insect conservation efforts in the Southern Hemisphere, therefore, have a heavy reliance on observing ongoing global changes and usually do

not have long-term historical data from which to draw comparisons (New, 2022). For example, the Rothamsted Insect Survey, which has been operated in the UK since 1968, allowed for the long-term population trends of large moths to be quantified over time showing that there was a spatially asymmetrical population decline in certain areas of the country (Conrad *et al.*, 2004). For other studies, population declines reported over longer periods of time would be undetectable or regarded as not significant if the populations were only monitored for shorter time periods (Hallmann *et al.*, 2017; Sánchez-Bayo and Wyckhuys, 2019). More data are required to truly understand how the insect populations of Southern Hemisphere regions, such as South Africa, are faring, and whether urgent conservation interventions may be necessary. Long-term studies are useful in gaining knowledge of insect diversity, although they are not ideal should rapid interventions be required (Weisser *et al.*, 2023). The continued threat of climate change creates urgency to increase understanding of how various aspects of biodiversity are expected to change over time. However, the number of studies that focus on changes of assemblages over time mainly focus on spatial gradients (Foord and Dippenaar-Schoeman, 2016).

The varied insect species found between the two latitudinal hemispheres on Earth have shown past trends of responding differently to climatic conditions and changes. For example, in cold climates, insects in the Southern Hemisphere have predominantly exhibited freeze tolerance, while those in the Northern Hemisphere utilise freeze avoidance strategies (Sinclair and Chown, 2005). The reason behind such differences is due to the different climate trends present in the Southern Hemisphere, where seasonal variation is reduced and climatic extremes are not as pronounced (Chown *et al.*, 2004). This is due to the Southern Hemisphere being mostly ocean-dominated instead of continental (Bonan, 2016). Therefore, the predictions and hypotheses proposed by most of the existing and established knowledge on conservation under climate change may not be fully appropriate to guide conservation decisions in South Africa. This poses an urgency to expand our understanding of current and future predicted changes in biodiversity and environmental conditions, and to create a localised conservation framework that will potentially be more successful in guiding conservation decisions.

Arthropods are relatively understudied when compared to more “charismatic” biological groups, and this disparity is especially exacerbated in developing countries which might not have the necessary resources allocated to research endeavours (Titley, Snaddon and Turner, 2017). Even in regions such as Europe, where there are far more economic and social resources available for research and conservation, only 0.12% of insect species have legal conservation protections (Leandro, Jay-Robert and Vergnes, 2017). Many laypeople are unaware of the value insects hold ecologically (Donkersley *et al.*, 2022). Therefore, they are not always considered and included when conservation strategies are implemented (van der Sluijs, 2020). However, there has been a growing amount of research and conservation interest in several insect groups. Charismatic species such as Odonata, Orthoptera, and Lepidoptera, as well groups that attract specialist attention such as some Coleoptera, have all been of greater focus in more recent years (Lewis, New and Stewart, 2007). Beetles have an importance in conservation due to factors such as their ecological diversity, presence as both pests and predators of pests, and presence in the fossil record, have all meant that they have been a useful group to analyse long-term faunal changes (New, 2007).

Beetles comprise the single largest order of animals globally, with an approximate 350,000 named species, and an estimated one million undescribed ones (Stork *et al.*, 2015). They also hold a prominent level of ecological influence due to their extreme ubiquity allowing for a considerable number of interactions with other forms of life that are present around them. Due to their vast ecological diversity, beetles provide a wide range of ecosystem services. For example, beetles facilitate many processes of decomposition, pollination, and seed dispersal (Kremen and Chaplin-Kramer, 2007; Oberhauser and Guiney, 2009). They are also especially useful bioindicators of environmental change (Koivula, 2011; Gonçalves *et al.*, 2021). Globally, beetles act as pollinators for hundreds of plant species, and are the sole specialists pollinating a majority of those species (Bernhardt, 2000). In the Cape Floristic Region, various species of Iridaceae have specialised to rely on pollination by Monkey Beetles (Family: Scarabaeidae, Subfamily: Hopliini) (Johnson, 2004; van Kleunen *et al.*, 2007). In another example, certain *Protea* species in KwaZulu-Natal were found to be most

often pollinated by flower chafer beetles (family: Scarabidae, subfamily: Cetoniinae) compared to other animal groups (Steenhuisen and Johnson, 2012).

Beetles act as detritivores by reducing litter, providing immense biomass for other organisms with a higher trophic level and also act as predators themselves (McKenna and Farrell, 2009). Macrodetrivores, such as tenebrionid beetles, are the main consumers of plant litter and are the primary organisms responsible for the decomposition and nutrient cycling of litter back into soil (Mordkovich and Afanas'ev, 1980; Cheli, Bosco and Flores, 2022). Thus, tenebrionids are essential ecosystem engineers and crucial for the maintenance of resilient environments and the trophic complexity found therein (Cheli, Bosco and Flores, 2022). Epigaeic carabid beetles specifically have been suggested to be potentially excellent bioindicators of climate change and ecosystem health due to their ecological and taxonomic diversity, and relative ease in sampling (Koivula, 2011). Carabidae also efficiently reflect their surrounding abiotic and biotic conditions due to their sensitivity to climate warming which can approach critical limits and tolerances. Carabidae exhibit observable responses to gradual changes in the surrounding environments with a shorter latency when compared to the vegetation which may be of primary concern (Brandmayr, 2016). Cicindelidae beetles that are also excellent bioindicators. Cicindelidae diversity has been shown to be usable as a proxy for butterfly and bird diversity present within the same areas but allow for faster sampling and assessment of population health than those groups (Pearson and Cassola, 1992). Regardless of their importance in the ecosystem or the services they provide, various beetles are threatened by climate change. Studies that examine the potential effects that climate change could have on the beetle species present in any respective area are essential in knowing the extent of the potential decline by climate change and what conservation actions might be necessary. Altitudinal gradients, and the varying conditions found across them, provide a proxy for how a species in a specific area might respond to changing environmental factors over time (Tito, Vasconcelos and Feeley, 2020).

Altitudinal gradients

Mountain habitats display disproportionately elevated levels of diversity concentrated into relatively small spatial extents. They also provide rapidly changing environmental factors with changing elevations, which allows for a variety of differing general environmental conditions to be present and affect the same populations of fauna and flora (Bishop and Robertson, 2024). This allows for observation on how certain taxa may respond to environmental changes and whether range shifts could be expected in response to factors such as climate change for other groups (Hodkinson, 2005). Specialist mountain beetle species have been shown to have greater risks of decline and eventual extinction when compared to those from other types of habitats such as forests (Nolte *et al.*, 2019).

One of the main determinants of insect presence and absence across altitudinal gradients are hypothesised to be the balancing of the thermal requirements needed to ensure optimal growth, versus a potential to surpass thermal tolerance limits (Birkett, Blackburn and Menéndez, 2018). Insect populations have been shown to shift into new altitudinal areas through altering their life histories and various phenological attributes in order to tolerate deviations from ideal environmental conditions in new habitats. However, any observed responses by insects to climate change across an altitudinal gradient will be species-specific (Hodkinson, 2005). Previous work on ants in Sani Pass, South Africa, concluded that seasonal fluctuations had a stronger effect on the distribution and density of ant species across the altitudinal gradient, in comparison to long-term climatic trends (Bishop *et al.*, 2014). However, long-term sampling is crucial in determining whether there is a change in the underlying climatic conditions as opposed to seasonal variation.

Insect diversity and abundance fluctuate in varying ways across altitudes depending on the specific circumstances involved. Climatic conditions, soil composition, biotic interactions, and individual species biology of the organisms involved all influence how the density of insect taxa will be spread across a mountain. Some examples of resulting distribution patterns include greatest richness and abundance surrounding mid-elevations, a linear decreasing relationship between diversity and elevation, and complex multi-modal patterns localised within altitudinal intervals (Dong *et al.*, 2025).

Environmental Changes and the Cape Floristic Region (CFR)

Soil variables and vegetation variables have been shown to have strong effects on terrestrial insects in historical studies (Duan *et al.*, 2016; Liu *et al.*, 2016; Liu, Wang, He, *et al.*, 2021; Liu, Wang, Wang, *et al.*, 2021; Cheli, Bosco and Flores, 2022). For example, Scarabaeidae exhibit a wide variety of manners in which they utilise soil and dung for reproduction. Telecoprids that rely on moving the dung ball around to house their eggs require a habitat with soil that is more easily manipulable (Daniel *et al.*, 2022). Changes in soil composition, such as increased clay concentrations cause the soil to become denser, therefore, beetles such as telecoprids may be unable to inhabit these environments, effectively altering their habitable range (Daniel *et al.*, 2022) (Farias and Hernández, 2017). This conclusion extends to any other groups of terrestrial insects that inhabit soil. Climate change will affect soil properties within a given environment (Bhengu and Onyeka, 2024), therefore resulting in pressure on the organisms both through direct thermal stress, and indirectly through altering their surrounding environments, making them unsuited for their ecological needs. For example, some tenebrionid species are restricted to sandy and loamy soil types due to their ecological behaviour and therefore would be affected by environmental changes that result in the soil they inhabit becoming less friable (Crawford, 1991).

The El Niño-Southern Oscillation (ENSO) is a protracted periodic extreme weather phenomenon. ENSO events have the potential to drive short-term changes and losses in diversity and function in insect populations (Sharp *et al.*, 2025). ENSO events also have the potential, due to their recurrence and intensity, to mask longer-term baseline trends of climate change that may be occurring in the relevant environments (Samways, 2010). The ENSO contributes to the challenge of improving our knowledge of insect populations in the Southern Hemisphere. Due to strong short-term climatic fluctuations, the observations of long-term climatic change in ongoing studies can be concealed (New and Samways, 2014). This is partially the reason the research and conclusions drawn from invertebrate studies in the Northern Hemisphere are not directly comparable and applicable to cases in the Southern Hemisphere (Sinclair and Chown, 2005).

Climatic oscillations such as the ENSO have altered the fire regimes in winter rainfall regions, such as the Cape Floristic Region (CFR) (Altwegg *et al.*, 2014). Other phenomena of atmospheric variability, one being the Antarctic Ocean Oscillation (AAO), also contribute to the changing fire conditions in the CFR (Wilson *et al.*, 2010). These changes occur due to the anomalous weather events, causing seasonal conditions that are drier and hotter than average. It is hypothesised that at the current rate of increased temperatures due to climate change, fire frequencies in these environments will substantially increase. This may lead to flora and fauna that are tolerant of more frequent fires to be the taxa that can most suitably inhabit these environments (Wilson *et al.*, 2010).

The CFR is the smallest and most species-rich floral kingdom globally (Taylor, Lee and Crous, 2001). Increasingly rising temperatures and reduced rainfall are predicted within the CFR, with its northern limits expected to face the greatest degree of change (Midgley *et al.*, 2003). The changes expected from the projected environmental changes, compounded with the additional threats of land use change and invasive species, are expected to cause the spatial extent of the CFR's vegetation to greatly contract (Midgley *et al.*, 2003). Most of the area threatened is predicted to occur in the northernmost boundaries of the Fynbos biome, mainly on the west coast and Cederberg mountains (Botes *et al.*, 2006) (Midgley *et al.*, 2002). In other global examples, focusing strategies to ecosystem-level monitoring (Vadas *et al.*, 2022), and conservation (Leandro, Jay-Robert and Vergnes, 2017) have been proposed as the most effective method to ensure the preservation of both the high levels of floral diversity and the fauna that are in the habitats they provide.

This study

Sampling of ground-dwelling insects was conducted in October of each year from 2002 to 2005, across an altitudinal gradient centred in the Cederberg Wilderness Area, in the northern Cape Floristic Region (Botes *et al.*, 2006, 2007). This current study repeated sampling the same sites 20 years after the original sampling. In the historical study, three beetle species were focused on due to their prevalence as the most abundant species overall in the transect (Gouws, 2007). Those species comprise the

carnivorous carabid, *Anthia decemguttata*, and the detritivorous tenebrionids, *Stenocara dentata* and *Zophosis gracilicornis*. The representation of differing feeding habits, ideal habitats, as well as representation across differing trophic groups make those species a compact but revelatory subset of the differing beetles in the study environment (Penrith, 1986; Picker, Griffiths and Weaving, 2019). Different trophic levels present across the entire ecological range within ecosystems have been shown to be varying sensitive to climate change effects, with the general trend that species of higher trophic levels are more sensitive (Voigt *et al.*, 2003). This proposed correlation has also been hypothesised to apply when comparing the trophic diversity between the insects of these ecosystems specifically (Pelini *et al.*, 2009). Studies have shown reduced feeding activity for detritivores under warmer and drier conditions (Thakur *et al.*, 2018), which are predicted climate change effects for this particular region (Midgley *et al.*, 2003). These hypotheses indicate that should the effects of climate change proceed as is, it could result in multi-level trophic impacts of the entomofauna assemblages present within the northern CFR.

Aims and objectives

In this study, I will address the following aims:

1. Investigate if any environmental conditions have changed across the study transect over the past 20 years.
2. From recent sampling, investigate the abundance and distribution of beetle assemblages across the transect.
3. Compare the distribution and abundance of three focal species (*Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilicornis*) to historical sampling.
4. Determine the environmental factors driving any observed changes.

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Chapter 2: Patterns of change in beetle assemblages over two decades

Introduction

Importance of insects, beetle assemblages, and factors influencing distribution

Insects provide a wide range of important ecosystem services globally, with pollination, natural pest control, decomposition, and ecosystem maintenance being only a few examples of the benefits they provide (Kremen and Chaplin-Kramer, 2007; Oberhauser and Guiney, 2009). It is estimated that in the United States, insects provide approximately \$60 billion worth of value annually (Losey and Vaughan, 2006). Beetles (Coleoptera) provide numerous ecosystem services that vary depending on their specific biological traits. Larger predatory and omnivorous families such as the Carabidae and the Cicindelidae regulate the populations of prey species, as well as engineering the environments around them by consuming the vegetative growths in the environment, potentially preventing any singular type of vegetation becoming dominant over others (Trichard *et al.*, 2014; Makwela, Slotow and Munyai, 2023). Macrodetrivores such as tenebrionid beetles provide the key function of maintaining healthy levels of soil fertility and nutrient cycling through decomposing plant litter and other organic matter (Mordkovich and Afanas'ev, 1980). Species such as *Nyctelia circumundata* (Tenebrionidae) in Patagonia have been observed to be key in maintaining the trophic structures of the environments in which they are active (Cheli, Bosco and Flores, 2022).

Beetles are one of the most diverse groups in the world with regards to both physiology and behaviour, therefore different groups of beetles will have their distributions affected by different factors (Liu *et al.*, 2016). There are three beetle families of focus in this study: Tenebrionidae, Carabidae, and Cicindelidae. Tenebrionidae are mostly detritivores while Carabidae and Cicindelidae are mostly omnivorous predators (Scholtz and Holm, 1985; Picker, Griffiths and Weaving, 2019). Historically, Cicindelidae had been considered a subfamily of Carabidae, and the taxonomic status of whether they are two distinct families has long been debated, although it has

recently reached a consensus where Cicindelidae were separated from Carabidae (Zhu *et al.*, 2025). However, due to the previous grouping and their similar ecologies, past and current ecological studies continue to examine them collectively as a unit in order to draw conclusions (Boutaud *et al.*, 2022). Hereafter any references to Carabidae implicitly includes both Carabidae and Cicindelidae. When considering epigaeic beetle groups with limited or no flight capabilities, they are affected to a much greater degree by environmental changes to their immediate surrounding areas as their dispersal is limited, meaning that the effects of anthropogenic climate change will be exacerbated for them (Homburg *et al.*, 2014; Duan *et al.*, 2016). The three beetle families mentioned prior all exhibit limited flight capabilities (Lövei and Sunderland, 1996; Carpaneto and Fattorini, 2002). Whether or not a species is affected by an environmental factor, and to what extent it is affected, is mostly dependent on the interaction between its mobility and trophic strategy (Liu *et al.*, 2016).

Various biotic and abiotic factors together determine beetle assemblage composition within their respective environments. These factors interact with the specific physiology and ecology of different insect species and determine whether those surroundings provide a suitable environment for them. This can either be due to environmental conditions that are not suitable for those species in question, or due to interspecific competition or predation by other animals present in the environment (Hodkinson, 2005).

Beetles have been shown to be affected by abiotic factors such as vegetation, soil and temperature. For vegetation some of the key variables include ground cover, height, vertical complexity, biomass, and diversity all relative to their specific areas (Liu *et al.*, 2016; Cheli *et al.*, 2021; Gonçalves *et al.*, 2021). For soil some of the key variables include pH, texture, moisture, salinity, organic content, and conductivity (Holland and Luff, 2000; Duan *et al.*, 2016; Cheli *et al.*, 2021; Liu, Wang, Wang, *et al.*, 2021). Temperature (air and ground temperature) was also shown to affect these families. For example, Carabidae were less affected by air temperature than ground temperature since they burrow to overwinter as larvae and adults (Lövei and Sunderland, 1996; Gonçalves *et al.*, 2021). For Tenebrionidae both air and ground temperature can affect their distribution to a great extent (Santos, 2002; Cheli *et al.*,

2021; Ren *et al.*, 2025). Light intensity and availability are related to vertical vegetative complexity but is still directly important for all the families and positively correlated with their distribution (Vodka and Cizek, 2013; Tyszecka, Zając and Kadej, 2025).

Precipitation is an important determining factor in the distribution of the Tenebrionidae and Carabidae. Tenebrionidae distribution is almost uniformly positively associated with increased levels of precipitation (Liu *et al.*, 2016; Cheli *et al.*, 2021). Several recent studies concluded that precipitation was the greatest determining factor in the distribution of the predatory beetle families (Liu, Wang, He, *et al.*, 2021; Liu, Wang, Wang, *et al.*, 2021). However, other studies in arid environments have shown the effects of precipitation differ with species being affected positively and negatively in the same study areas in response to changing precipitation (Tsafack *et al.*, 2019). For Carabidae there are also a number of other factors that are specifically highly effective at being a predictor for their distribution, such as abundance of prey, and interspecific competition with other predators (Niemelä, 1993; Lövei and Sunderland, 1996; Gonçalves *et al.*, 2021).

Altitudinal interactions with assemblages and environmental factors

Studying altitudinal gradients is an effective strategy for understanding the distribution ranges and environmental tolerances of several types of organisms. This is because the altitudinal gradient allows for a wide range of environmental variables, vegetation composition, and the maximal and minimal limits of the species in question to those variables to be captured and analysed across relatively small spatial extents (Menéndez *et al.*, 2014; Rutten *et al.*, 2015). Different mountains will have completely unique histories involving specific environmental and geographical intricacies, changes in vegetation composition across the elevation gradient, as well as particular combinations of species interacting and forming assemblages. These factors all mean that mountains are usually biodiversity hotspots with high levels of endemism inhabited by a variety of specialist species (Soh *et al.*, 2019; Musthafa *et al.*, 2021). Mountains are also useful for conservation studies in this era of anthropogenic climate change because they allow for the importance of temperature for specific organisms to be studied observationally. Most mountainous environments exhibit a consistent

inverse relationship between altitude and elevation (Röder *et al.*, 2017; Vetaas, 2021). Any responses shown by insects towards temperature gradients across an altitude, could therefore be used as an analogue for climate change and increasing temperatures over time in a fixed location (Hodkinson, 2005; Bishop and Robertson, 2024). Therefore, studying how certain taxa are distributed and affected by the conditions across a mountain allows for a crucial baseline where the current and expected responses of local biodiversity can be continuously monitored (Botes *et al.*, 2006; Bishop and Robertson, 2024). For example, the predicted response of many ectotherms such as insects towards increasing temperatures is an expansion of their maximal altitudinal range (Wilson *et al.*, 2005). It is expected that environments at higher elevations might have been previously unsuitable for certain taxa due to the temperatures being below their minimal thermal limits (Wilson *et al.*, 2005; Menéndez *et al.*, 2014). However not every group will react in this specific way.

An upwards shift in distribution has been observed in Geometer moths (Geometridae) on a tropical mountain in Malaysia (Chen *et al.*, 2011). Various dung beetle species have also shown up-slope range shifts that were attributed to increasing temperatures (Menéndez *et al.*, 2014). Vegetation is also affected by the environmental gradients present across a mountain, with most of the vegetation having shifted upward in their elevational range over the last century due to climate change (Lenoir *et al.*, 2008). However, recent studies have shown that as global temperatures have increased, other factors such as changing precipitation regimes and intensifying wind speeds have also meant that some plants are also shifting downwards in their range (Zu *et al.*, 2023). Due to high levels of specialisation shown on mountain ecosystems, there are several beetle taxa, most commonly herbivorous species, that have a dependent relationship on specific host plants for their food and habitat. Those species have altitudinal ranges that are closely tied to their host plants and will usually shift in response to changes in the plant's distribution rather than other environmental variables (Hodkinson, 2005). Certain insects might rely on host plants through tolerating the ambient temperature of their environments by using the microclimate it provides in areas they otherwise would not be able to inhabit (McCain and Garfinkel, 2021).

Environmental changes

Climate change is one of the most prevalent causes behind the mass decline in entomofauna abundance and biodiversity that is being observed globally (Forister, Pelton and Black, 2019; Sánchez-Bayo and Wyckhuys, 2019; Harvey *et al.*, 2023). The most recent IPCC Synthesis Report discussed that immediate interventions were necessary to reduce greenhouse gas emissions to meet what was previously considered a worst-case scenario of a 1.5°C global temperature increase and avoid the now projected 2°C global temperature increase projected for 2050 (Calvin *et al.*, 2023).

Insect populations will have various responses to the effects of long-term anthropogenic climate change. The baseline increase of mean global temperatures will mean that different taxa living in specific environments will gradually experience their suitable ranges shrinking due to other areas increasingly being above their critical thermal limits. Associated changes in other important factors such as host vegetation availability and soil moisture reduction, will result in less habitable space for most insects, and will usually result in mass mortality for many populations (Harvey *et al.*, 2023). The extent of the loss in insect distribution, biodiversity, and abundance will depend on the severity of climate change impacts over the coming decades. Scientists have stated the importance of insects in ecosystems globally with respect to the ecosystem services they provide. Many of those insects are declining in abundance and many have already become extinct, with an estimated 70% reduction in flying insect abundance over the last 30 years, and 60% reduction in large Carabidae in the United Kingdom over a 15-year period (Brooks *et al.*, 2012; Cardoso *et al.*, 2020).

In Southern Africa, the effects of increasing global temperatures are expected to be droughts, increased heatwaves, extreme loss of soil moisture, with all these factors contributing to more frequent and intense fires throughout the region (Midgley *et al.*, 2003; Hannah *et al.*, 2005). This general drying of the environment is expected to affect both summer and winter rainfall biomes (Harvey *et al.*, 2023; Munday *et al.*, 2025; Sharp *et al.*, 2025). The effects of climate change are also expected to cause disruptive climatic phenomena such as the El Niño Southern Oscillation (ENSO) to increase in frequency as well as intensity (Wang *et al.*, 2019). These changes in the

expected behaviour of the ENSO have been suggested to be strongly linked to anthropogenic greenhouse gas emissions (Cai *et al.*, 2023). ENSO also confounds observations made to record baseline climate, providing an additional layer of difficulty in analysing the extent of climate change and its potential effects in respective areas. This means that more frequent and intense ENSO events will also make future monitoring for conservation priority guidance more difficult (Samways, 2010).

Background and Aims

This study aimed to compare beetle assemblages to time series sampling conducted 20 years ago (2002-2005) across an altitudinal gradient in the Cederberg Wilderness Area in the northern Cape Floristic Region (Botes *et al.*, 2006, 2007). This sampling was repeated during 2022-2024, utilising the exact same sites and sampling configuration. Various environmental and climatic factors such as proportionate ground cover, vegetation heights, continuous temperature readings, and soil samples, were collected and compared to historical environmental data that was collected during the original study (Botes *et al.*, 2006). In particular, my main aims were to: 1) investigate if any environmental conditions have changed across the study transect over the past 20 years; 2) investigate the abundance and distribution of beetle assemblages across the transect from recent sampling; 3) from recent sampling, compare the distribution and abundance of three focal species (*Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilicornis*) to historical sampling; and 4) determine the mechanisms behind any observed changes.

I hypothesised that there will be an increase in the abundance of beetles at higher altitudes compared to the historical study, due to the expected increase in mean temperature across the entire transect (Wilson *et al.*, 2005).

Methods

Site and sampling

The methodology and sampling of this study are identical to those used in a historical study started in 2002 (Botes *et al.*, 2006, 2007). This study was conducted in the Greater Cederberg Biodiversity Corridor, specifically across an altitudinal transect that spans both aspects (East and West) of the Cederberg mountains. This altitudinal transect comprises the major vegetation types found within the Cederberg Wilderness Area, being Strandveld Succulent Karoo, Mountain Fynbos, and Succulent Karoo (Mucina and Rutherford, 2006), therefore representing almost the full range of vegetation that could be expectedly observed in this region.

A total of 17 different sites were sampled with 200 metre altitudinal intervals between each, ranging from sea level at Lambert's Bay on the western aspect of the transect (sea level, Strandveld Succulent Karoo), to Sneeuwkop (1926 metres a.s.l.), and continuing down the eastern aspect of the mountain and ending near the town of Wupperthal (Lowland Succulent Karoo, 500 metres a.s.l.) (Figure 1). All the sites between the two endpoints of the transect were representatives of Mountain Fynbos vegetation. These Mountain Fynbos sites were categorised as Restioid, Ericaceous, and Proteoid (Mucina and Rutherford, 2006) according to the respective dominant vegetation family found within that environment as also determined by Botes *et al.* (2006, 2007).

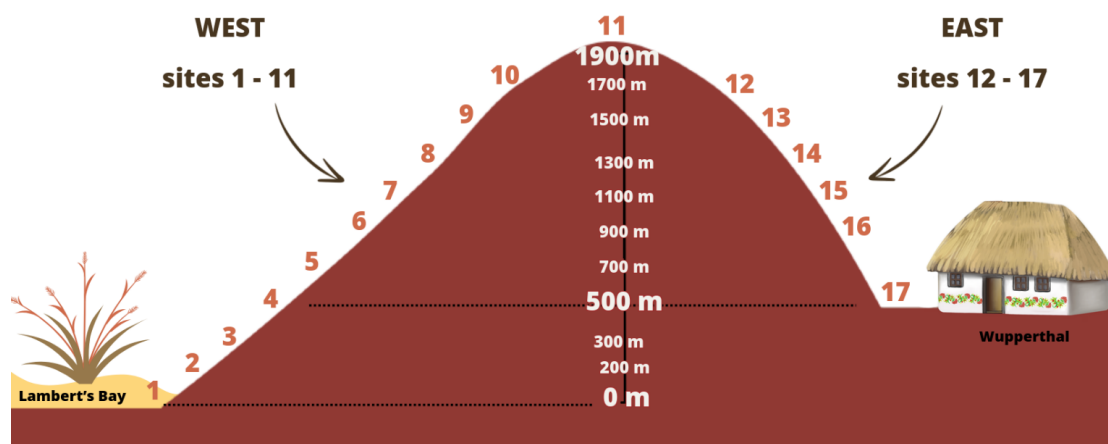


Figure 1: Diagram representing the study transect.

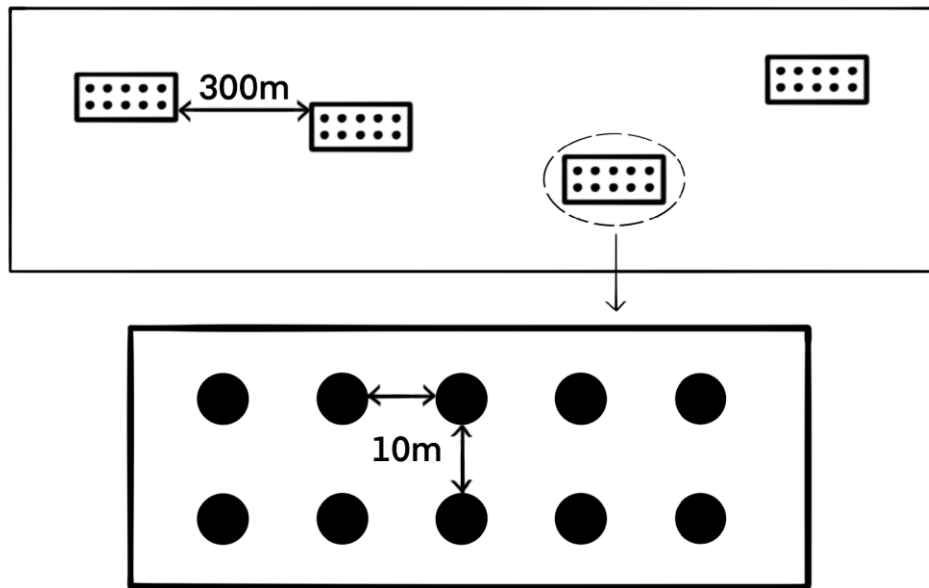


Figure 2: Sampling configuration (adapted from Botes *et al.*, 2006).

Pitfall traps were used to sample epigaeic insects across the transect. At each of the 17 sites four areas were chosen as replicates, with each replicate being at least 300 metres away from any of the other replicates. At each of these replicates ten pitfall traps were buried in the soil in such a way that the rim of the pitfall trap was level with the ground. The present vegetation was not cleared away when placing the pitfall traps and disturbance of the surrounding areas was kept minimal. Each of the ten pitfall traps were placed in a grid made up of two rows containing five traps each, with each trap being at least 10 metres away from any other trap (Figure 2). GPS coordinates were recorded for every trap for use with spatial analyses later. The traps were half-filled with a 99.9% propylene glycol solution and were left out for five nights before being collected. In the historical study, beetles were sampled in October 2002 and October 2003 (Botes *et al.*, 2006, 2007). Thereafter, sampling was undertaken every October and March from October 2022 until March 2024. For the purpose of this study, the October 2022, March 2023, and October 2023 samples were used.

During October 2002 and 2022, a 20cm³ soil sample was collected from each replicate by taking 20 sub-samples of soil randomly across replicate sampling area, then these sub-samples were homogenised, and the final sample was taken from this mixture. After the soil samples were air-dried in the laboratory for 10 days, they were sent to BemLab (Pty Ltd.), Strand, South Africa to be analysed to determine the proportional composition, mineral contents, ionic contents, pH, and conductivity.

Vegetation was sampled in two different ways at each trap to determine both horizontal and vertical distribution of the vegetation. The ground cover of vegetation distribution was determined by placing a 1-m² quadrat down with the pitfall roughly at the centre. Following that, the percentage of the ground that was covered by the four classifications of leaf litter, live vegetation, exposed rock, and bare ground were estimated. After estimating the ground cover, a 1.5-metre-long rod was placed at each corner of the quadrat. This rod was marked at 25 cm increments and the number of times vegetation made contact with the rod within each height interval was recorded. These counts of contact were used to determine foliage height profiles surrounding each trap. In the laboratory, beetles were identified to morphospecies level based on their external morphological characteristics (Barratt *et al.*, 2003). Each beetle was either pinned or mounted depending on its size, and they were then labelled and grouped by sample. The 2022-2023 beetles were identified by a taxonomic expert (Werner Strumpher, Ditsong National Museum of Natural History). Beetles belonging to the Tenebrionidae, Carabidae, and Cicindelidae families were used for all analyses (historical vs recent sampling), as these are the same three families that were analysed in the historical study (Botes, McGeoch and Chown, 2007). In addition, three focal beetle species—the carabid *Anthia decemguttata* (previously *Thermophilum decemguttatum*) (Figure 3), and the tenebrionids *Stenocara dentata* (Figure 4), and *Zophosis gracilicornis* (Figure 5)—were determined to be the three most abundant and widespread beetle species present across the transect in the current and historical studies, thus were used for historical comparisons



Figure 3: Photograph of pinned *Anthia decemguttata* specimen.



Figure 4: Photograph of pinned *Stenocara dentata* specimen.



Figure 5: Photograph of pinned *Zophosis gracilicornis* specimen

Climate modelling and environmental data

Soil temperatures across the transects were measured hourly over the 20-year duration of this study using iButton sensors (Semiconductor Corporation, Dallas/Maxim, TX, USA). Temperature observations were halted from late 2020 to 2022, thus a mechanistic model was utilised to model the microclimate temperatures present. The “micropoint” model was used to model the ground-level soil temperatures present within the sampling area (Maclean, 2025). This mechanistic model required the input of the meteorological data of the locations to be modelled, as well as defining parameters that correspond to the vegetation, soil, and ground in those areas. Spatial meteorological data were obtained using the MCERA5 package in R (Klinges *et al.*, 2022), which is based on the ERA5 global climate re-analysis (Hersbach *et al.*, 2020). Using the GPS coordinates of each replicate at every altitudinal site across the transect, hourly observations from 2002 to 2024 were extracted for every sampling replicate. The climate variables of air temperature, relative humidity, atmospheric pressure, shortwave radiation, longwave radiation, diffuse radiation, wind speed, wind direction, and precipitation were used to run the microclimate model (Maclean, 2025). Data of previous ENSO events were used to create a plot showing the timeline of El Niño and La Niña events over time (National Oceanic and Atmospheric Administration, 2026).

Soil was analysed at each sampling site for their proportional composition of silt, clay, sand, and stone. These compositional measurements were used to group the soil at each site into distinct categories. These classifications were utilised to apply the corresponding pre-set soil parameters for model creation provided by the micropoint package. The parameters necessary for the model were ground reflectance, ground emissivity, soil bulk density, volumetric mineral fraction of soil, volumetric quartz fraction of soil, mass fraction of clay, a Shape parameter for Campbell soil moisture models, matric potential of soil, volumetric water content of soil at saturation, residual water content of soil, a shape parameter of the van Genuchten model, pore size distribution parameter for the soil, and the saturated hydraulic conductivity of the soil (Maclean, 2025). Other required ground parameters of slope and aspect were determined using the NASA SRTM digital elevation model (NASA JPL, 2013).

The vegetation parameters required for the microclimate model were determined using the average vegetation cover and vegetation height that were measured at each pitfall trap, averaged across replicates. These measurements were utilised to generate the required parameters of canopy height, plant area index, leaf inclination angle, canopy clumping, leaf reluctance of shortwave radiation, leaf transmittance of shortwave radiation, the diameter of leaves averaged over both the length and width of leaves, emissivity of vegetation, maximum stomatal conductance, and stomatal sensitivity (Maclean, 2025). Using all the required data and generated parameters, the hourly microclimate temperatures at both the soil level and 0.5cm above the ground were calculated using the micropoint model. These modelled temperatures were compared graphically to the empirical temperature observations of both the iButtons and the meteorological temperature observations. The meteorological data were also utilised to extract the hourly precipitation over time per replicate along with the temperature modelling. All these data were extracted and analysed using R and the “MCERA5” package (Klinges *et al.*, 2022).

Utilising the CapeNature database (Forsyth, 2023), a list of historical fires that had occurred within the bounds of the study area were compiled. The extent of each fire and its date were stored using spatial polygons through the “sf” package in R (Pebesma and Bivand, 2023). The fires were attributed to each pitfall trap using the GPS coordinates of the trap. Through cross-referencing the site coordinates and the extent of the polygons, a map and database was created which showed the list of all the fires a particular site had experienced in recorded history, as well as the most recent fire, and consequently the years since the last fire. These fire data were used in conjunction with the other environmental data to provide a further explanatory variable to explain the environmental changes that were observed.

Statistical analyses

All data analyses, except for the calculation of slope and aspect based on GPS coordinates, were performed in R V.4.5.3. (R Core Team, 2026). For the October 2022 and March 2023 samples, all the beetles were compared to each other visually using bar charts and Non-metric Multidimensional Scaling (NMDS) plots. The abundance

and species richness between the two seasons were also compared quantitatively, across the entire transect and on a site-by-site basis. The abundances of the three focal species (*Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilicornis*) were compared between the recent collections of these three species (October 2022, October 2023, and March 2023) and the historical collections (October 2002 and 2003). Although diversity could not be compared directly between the current sample and the historical sample, species density could be used as a comparison as it does not rely on the same species to be matched or identified in the different samples. Species density is calculated as the number of species present per unit area in the sample. This measure of species density was also used to compile sample-based rarefaction curves for each site, across all the identified beetle species from the 2022-2023 sample, to determine the degree of sampling representation (Gotelli and Colwell, 2001). These rarefaction curves were made using the “vegan” package in R (Oksanen *et al.*, 2025).

Redundancy analysis (RDA) was used to examine the relationships between the beetle sampling and assemblage structure with the environmental variables (Oksanen *et al.*, 2025). RDA was performed on the historical beetle comparisons, the current seasonal comparisons, as well as the October 2022 sampling season. For the current samples where all the species from the three families of interest were identified, both the abundances, and the presence-absence data of the beetles were analysed with RDA.

The results of the significant variables obtained from the RDA, as well as prior known explanatory factors that affect beetle abundance and diversity, were used to construct generalised linear models (GLMs) to quantify the effects that these various factors may have on the beetle assemblages. Specifically, due to the sampling being repeated on the same locations across a temporal gradient, Generalised Linear Mixed-Effects Models (GLMMs) were used (Hadfield, 2010; Bates *et al.*, 2015).

Whether or not beetle species were shifting in their altitudinal range were analysed. The data were compared both using all the species compared between the current beetle samples, as well as the three focal species compared between the current samples and the historical samples. The Jaccardian similarities of the different sites

compared between the current and historical sampling efforts were computed to show the differences in beetle assemblage composition within each site across the 20-year gap between sampling.

The environmental variables of vegetation ground cover, vegetation height, and soil attributes, were also compared across the various temporal categories corresponding to the different beetle sampling events. They were compared to determine whether the environment surrounding the sampling sites had changed significantly between the different sampling efforts. The changes in environmental conditions combined with the effects that those variables have on beetle assemblage responses as shown in the GLMs will be used to determine whether those changes could be driving reasons behind the population changes. The fire data were used to determine whether there was a relationship between the recorded fire, and the environmental variables present within the sampling sites.

The time-series data for the modelled microclimate temperatures (air and ground) as well as the precipitation observations were analysed using Seasonal-Trend Decomposition with Regression (STR) (Dokumentov and Hyndman, 2022). The STR models separate the changes in the response variable into parts of cycling seasonality (daily, yearly, etc.), and trend over time. This allows for baseline trends in the time-series data to be extracted and determine if there are any potential observable directional changes over time. After determining any potential trends per site, descriptive representatives of the relevant variables can also be incorporated in the interpretation of assemblage changes over time and also utilised in any models.

Results

Environmental variables

The proportional composition of ground cover was observed to have changed significantly between the historical and current sampling events. There has been a general trend where the proportion of the bare ground and exposed rock has decreased, while the proportion of plant litter and live vegetation cover have all proportionally increased (Figure 6).

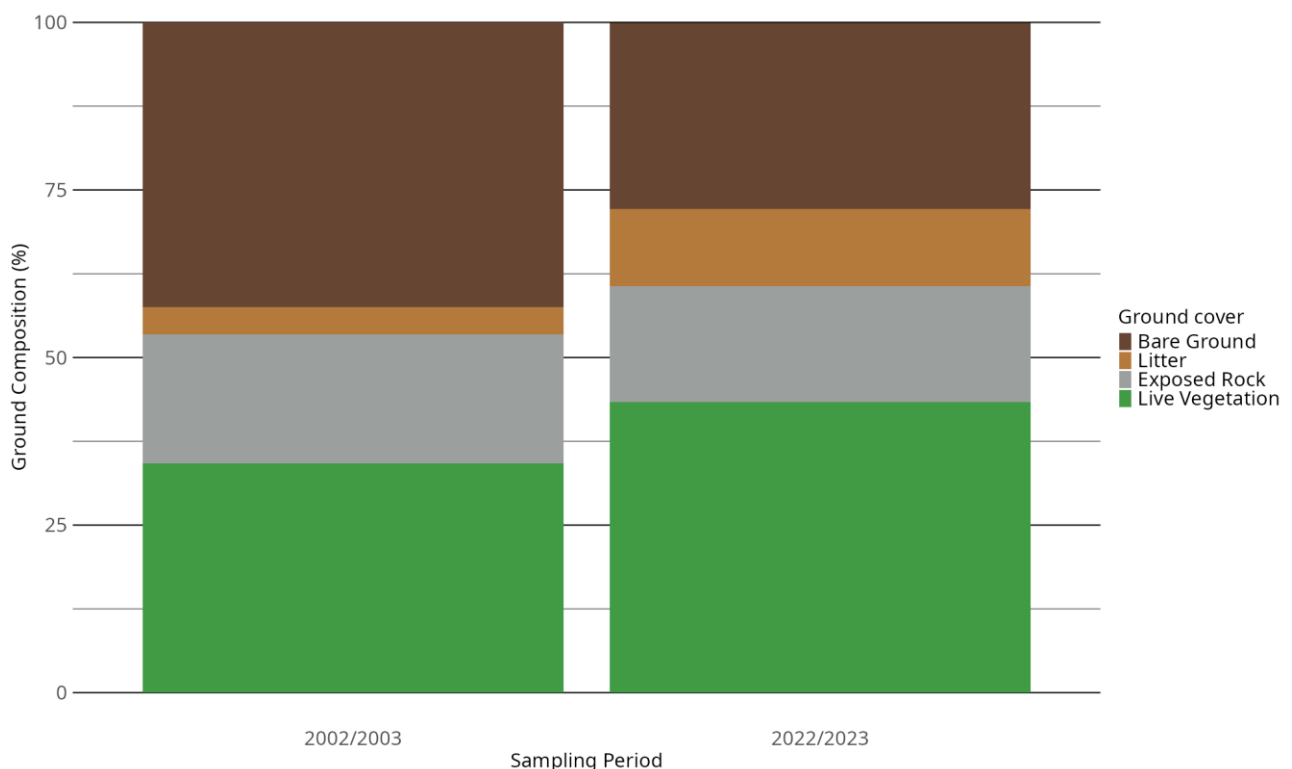


Figure 6: Mean proportional composition of the ground cover present across the entire transect, compared between the two different sampling periods.

When these data were analysed with the GLMMs, all these changes in proportional composition were determined to be significant. Across the time that has passed between the two sampling periods, bare ground has decreased by 35% (42.5% in 2002–2003, 27.8% in 2022–2023), plant litter has increased by 183% (4.07% in 2002–2003, 11.5% in 2022–2023), exposed rock has decreased by 10% (19.3% in 2002–

2003, 17.3% in 2022–2023), and live vegetation cover has increased by 27% (34.2% in 2002–2003, 43.4% in 2022–2023), across the entire transect (Table S1).

Vegetation height has changed significantly over time across the entire transect, with both the mean vertical complexity of vegetation and the mean maximum vegetation height increasing in the time between the sampling periods (Figure 7). When analysed with GLMMs these changes were determined to be significant, with mean vertical complexity of vegetation increasing by 28% (8.41 in 2002–2003, 10.7 in 2022–2023) and mean maximum vegetation height increasing by 54% (29 cm in 2002–2003, 58 cm in 2022–2023) (Table S2). The changes in these vegetation height characteristics were site-specific, and further GLMM analysis including site as an explanatory variable showed in which sites there were significant changes.

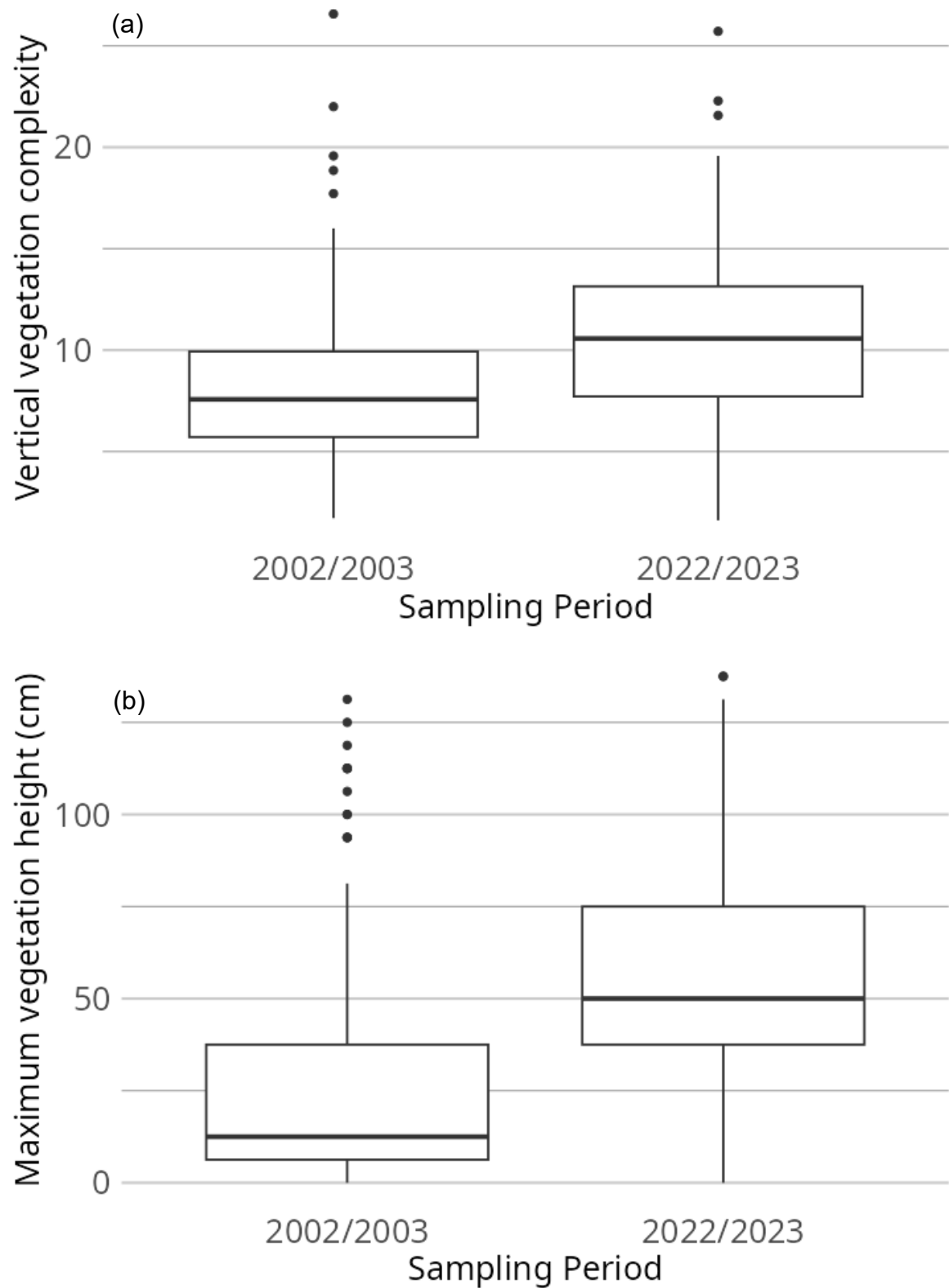


Figure 7: The (a) vertical vegetation complexity, and (b) mean maximum vegetation height, across the entire transect compared between the different sampling periods.

Temperature and precipitation seasonal decomposition

The comparisons between the iButton sensor data and the modelled microclimate data showed that the temperatures were matching overall and were therefore valid replacements for the empirical temperature observations and the missing gaps of temperature monitoring contained therein (Figure S1).

The STR models generated for each individual site were examined to see if their trend components contained any noticeable change over time. There was a general increase in both air and ground temperatures across the entire transect over time (Figure 8). Keeping seasonal fluctuations and climatic phenomena in mind, there has been a 0.19°C ($t = 878.289$, $P < 0.001$) and 0.23°C ($t = 727.2$, $P < 0.001$) increase in the baseline air and ground temperatures, respectively.

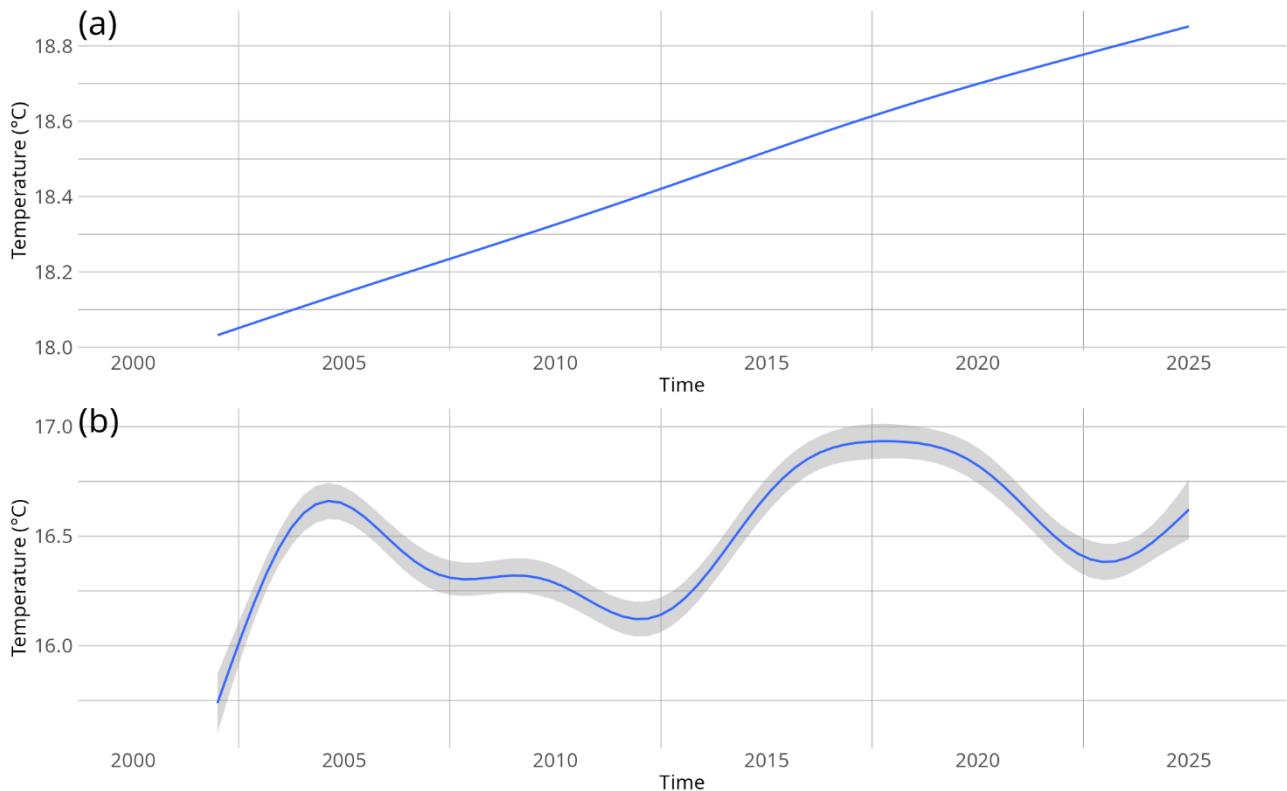


Figure 8: Temperature trend over time for (a) ground temperature, and (b) air temperature across the entire transect.

Some of the sites exhibited a visibly monotonic temperature trend over time, for example the site at 700 metre altitude on the western slope (Figure 9). This temperature trend was quantified at an estimated 0.84°C ($t = 210.8$, $P < 0.001$) increase in ground temperature, and a 0.19°C ($t = 1231.212$, $P < 0.001$) increase in air temperature over the 23-year period. A total of eight sites across the transect showed this trend of monotonic baseline temperature increase over time (Figure S3): "500m(W)", "700m(W)", "1100m(W)", "1300m(W)", "1700m(E)", "1300m(E)", "900m(E)", and "500m(E)".

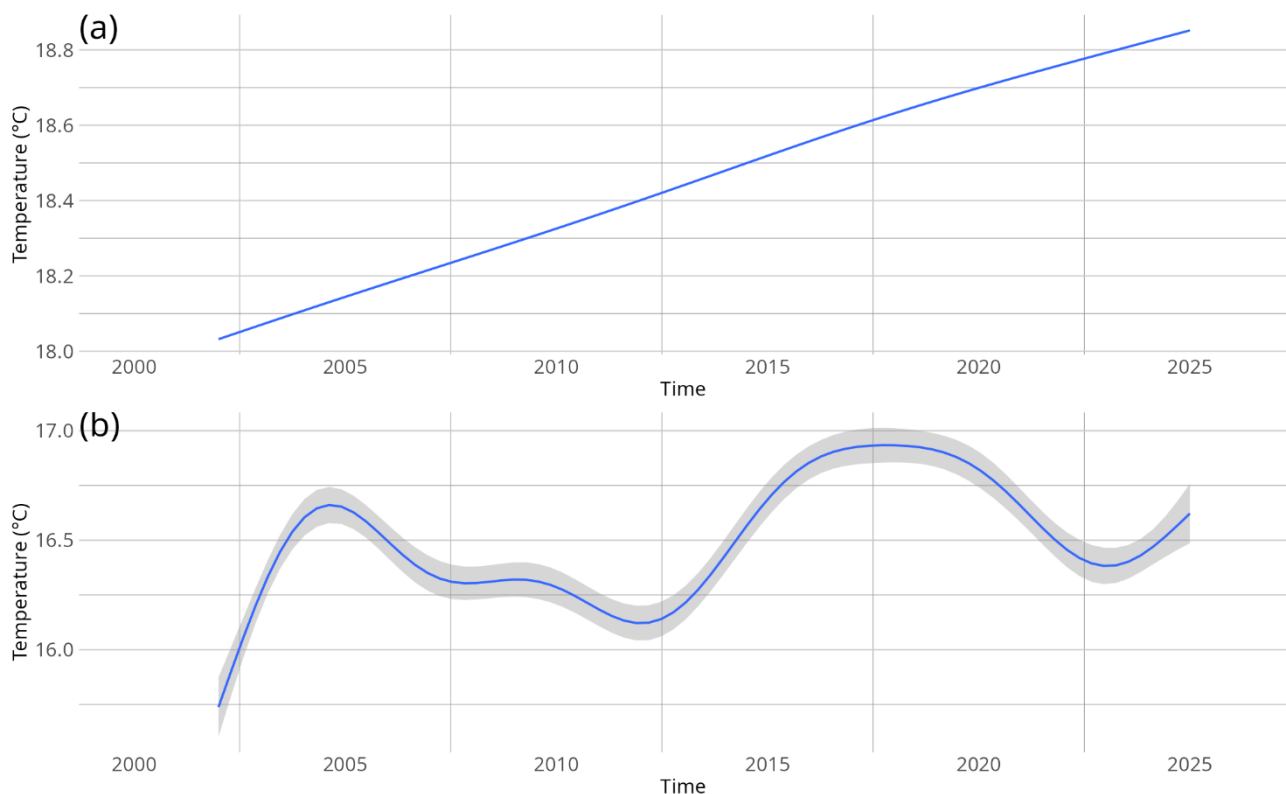


Figure 9: Temperature trend over time for (a) ground temperature, and (b) air temperature for the site at 700m altitude on the western slope.

There was a consistent change over time, even in sites where the baseline temperature did not display a trend, of the cyclical seasonality of air temperature becoming increasingly more extreme in terms of both local temperature minima and maxima (Figure 10). This general observation both across the entire transect and

within each individual site, with varying degrees of increased extremity in the seasonality.

The seasonal decomposition plots for precipitation did not show any monotonic changes in the precipitation trend over time for any particular sites. However, there was an overall cyclical trend change where there have been multiple peaks and troughs of baseline precipitation over the years (Figure 11a). Most of the variation in precipitation is observed in the first five sites of the transect ("0m(W)", "200m(W)", "300m(W)", "500m(W)", "700m(W)"), with the remaining twelve sites having an almost identical precipitation history across the history of the study (Figure S2). Despite there not being a monotonic change in the baseline trend, there was a consistent change in the cyclical seasonality of precipitation, exhibiting steady increases in both the minimum and maximum precipitation present within the cycle over the years (Figure 11b). The extent of the increasing precipitation seasonality differed by altitude, with some showing a constriction in the seasonality range.

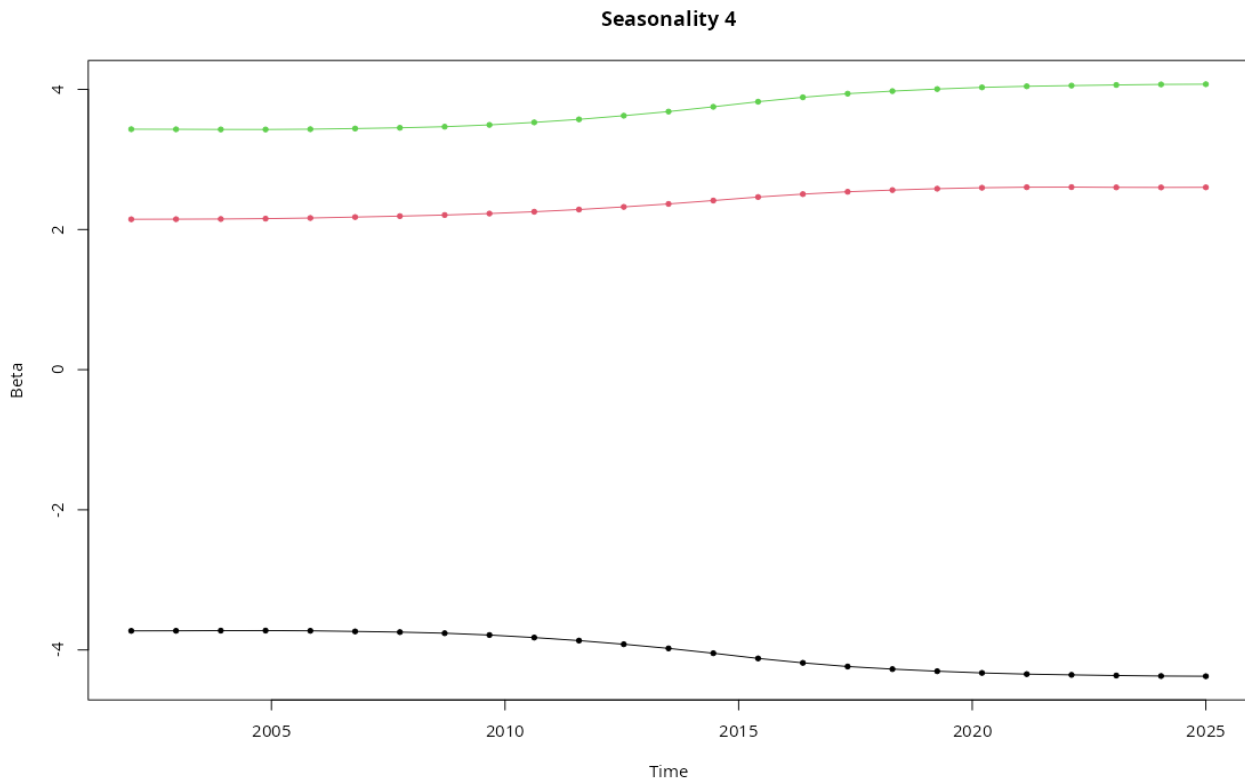


Figure 10: Seasonality of air temperature for the entire transect, showing the change from 2002-2025 of the coldest time of the day (bottom black line), hottest time of the day (top green line), and the average temperature of the day (middle red line).

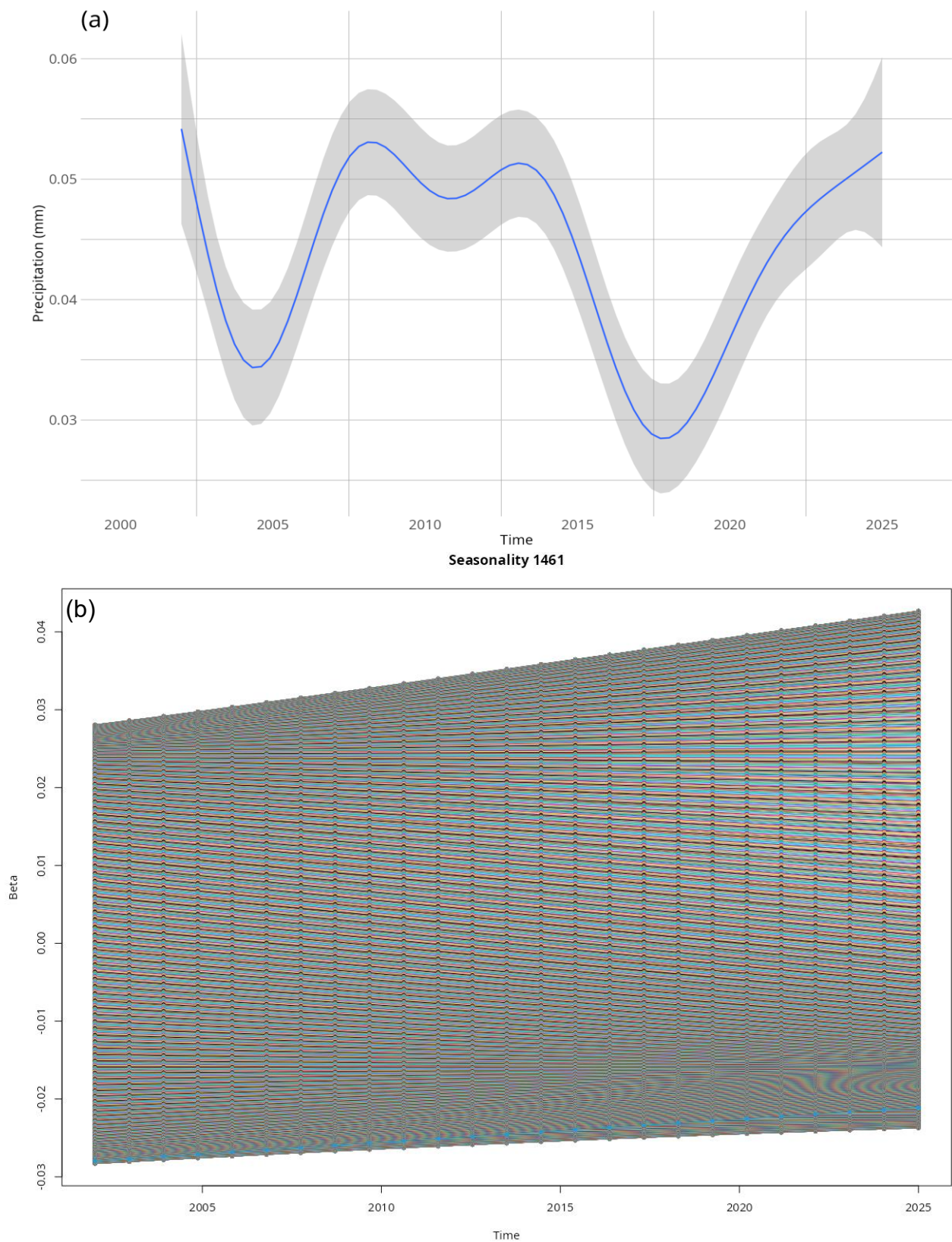


Figure 11: Trend in (a) baseline precipitation, and (b) seasonal precipitation range over time, for the study transect over time.

Current beetle assemblage and seasonal comparison

From recent sampling (October 2022 and March 2023), a total of 42 beetle species were collected from the three families that were being studied, with a total of 1,633 individual beetles collected (1,589 individuals from October 2022 and 44 individuals from March 2023) (Table S14). Thirty-four of the species belonged to the Tenebrionidae, six to Carabidae, and two to Cicindelidae. Forty-two species were present in the October 2022 samples, with only five of these species present in the March 2023 samples (*Anthia decemguttata*, *Stenocara dentata*, *Zophosis gracilicornis*, *Cryptochile* sp. 1, *Gonogenia* sp. 1). The highest mean abundance was observed in the Ericaceous Fynbos, with the Succulent Karoo site having the second highest mean abundance, and the Proteoid and Restioid Fynbos sites having similar mean abundances. The lowest mean abundance was observed in the Strandveld and Alpine Fynbos sites. The mean species richness was similar across all the different vegetation types, with the Proteoid Fynbos sites having the highest richness (Table 1). For the March 2023 samples specifically, the highest richness and mean abundance were observed in the Succulent Karoo site (Figure 12).

Although no general trends were observed for species richness and abundance across the altitude, there was a distinct abundance peak situated at the “1100m (W)” site (Table 2). Also, one site “1300m (E)”, exhibited a different assemblage structure and composition when compared to all other sites in the general slope trend across both sides of the transect. This site was very homogeneous and was dominated mostly by one single species (*Cicindela lurida*), which resulted in a site with relatively high abundance compared to its diversity (Figure 13).

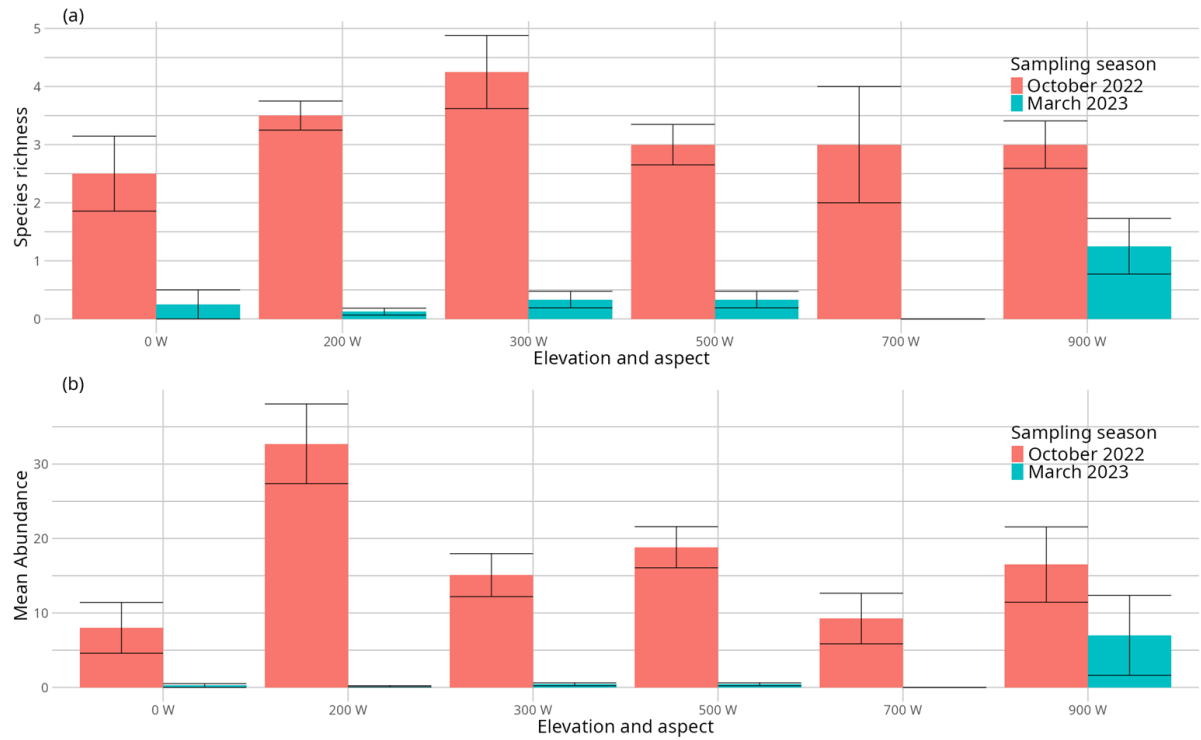


Figure 12: (a) Species richness ($\pm$ standard error), and (b) mean abundance ($\pm$ standard error) across the different vegetation types present in the transect, separated by season.

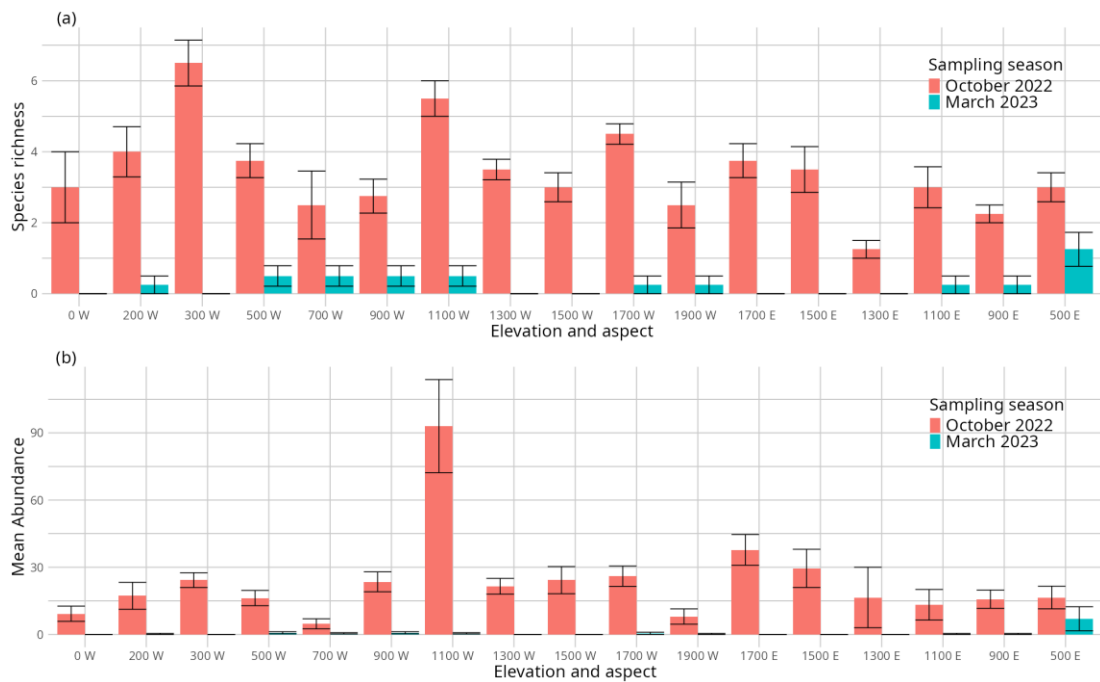


Figure 13: (a) Species richness ($\pm$ standard error), and (b) mean abundance ($\pm$ standard error) across the altitudinal gradient of the transect, separated by season.

Some of the species shifted in their altitudinal range between seasons, with the range shifts differing between the two aspects of the transect. On the western slope, the lower elevational limits increased for four different species, and two of those species also had a decrease for their upper elevation limits (Figure 14a). On the eastern slope, three species experienced a downwards contraction of their upper elevation limits, while one species experienced an upwards contraction of its lower elevation limits (Figure 14b).

Comparing the abundance between the seasonal samples, the NMDS plot showed that aside from a few outliers, the samples from the two different seasons showed an overlap (Figure 15a). The comparison of species presence-absence showed that while the October 2022 assemblage had many sites with similar species being present, the different observations for March 2023 did not show any signs of grouping (Figure 15b).

The NMDS plots comparing species abundance and presence between the different altitudinal sites showed that most of the altitudinal sites were similar with only a few notable exceptions (Figure S4). Site 11 (the summit, 1900m), site 14 (1300m (E)), and site 1 (Lambert's bay, 0m (W)) were all distinctly different from the general grouping shown by the other sites.

Comparing the species abundance and presence between the various categories of vegetation groups in the sites showed that the different subgroups of Mountain Fynbos were all more similar to each other than the other vegetation types. The Proteoid and Restioid sites were the most similar with the Ericaceous sites only differing slightly. The Succulent Karoo site was also similar to the different Fynbos groups but much more closely clustered than the Fynbos sites. The Alpine and Strandveld sites were both very dissimilar to each other and the other groups previously mentioned. The Alpine site was especially very varied within the site itself (Figure 16).

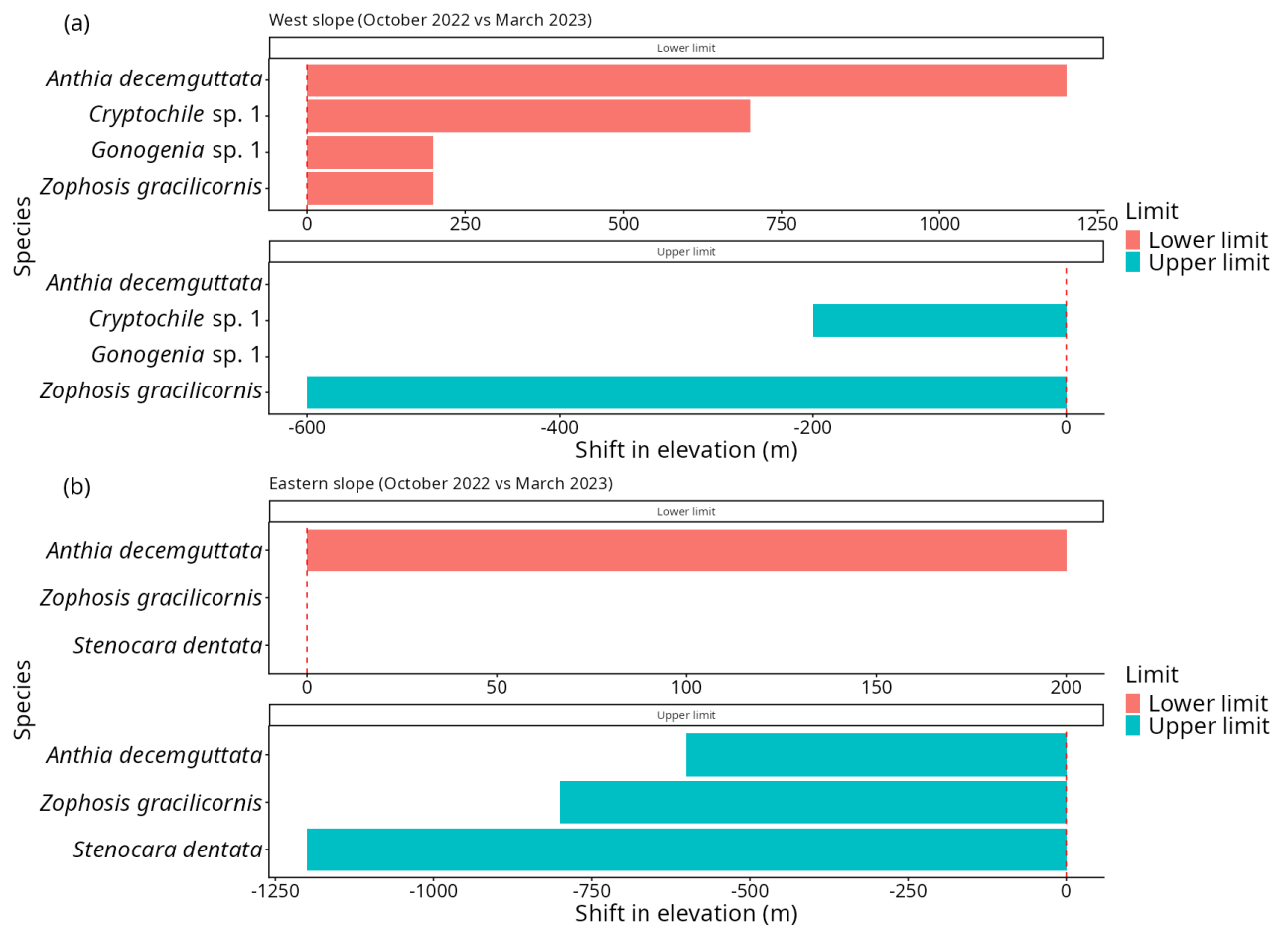


Figure 14: (a) Western slope, and (b) eastern slope altitudinal range limit shifts for relevant species between October 2022 and March 2023.

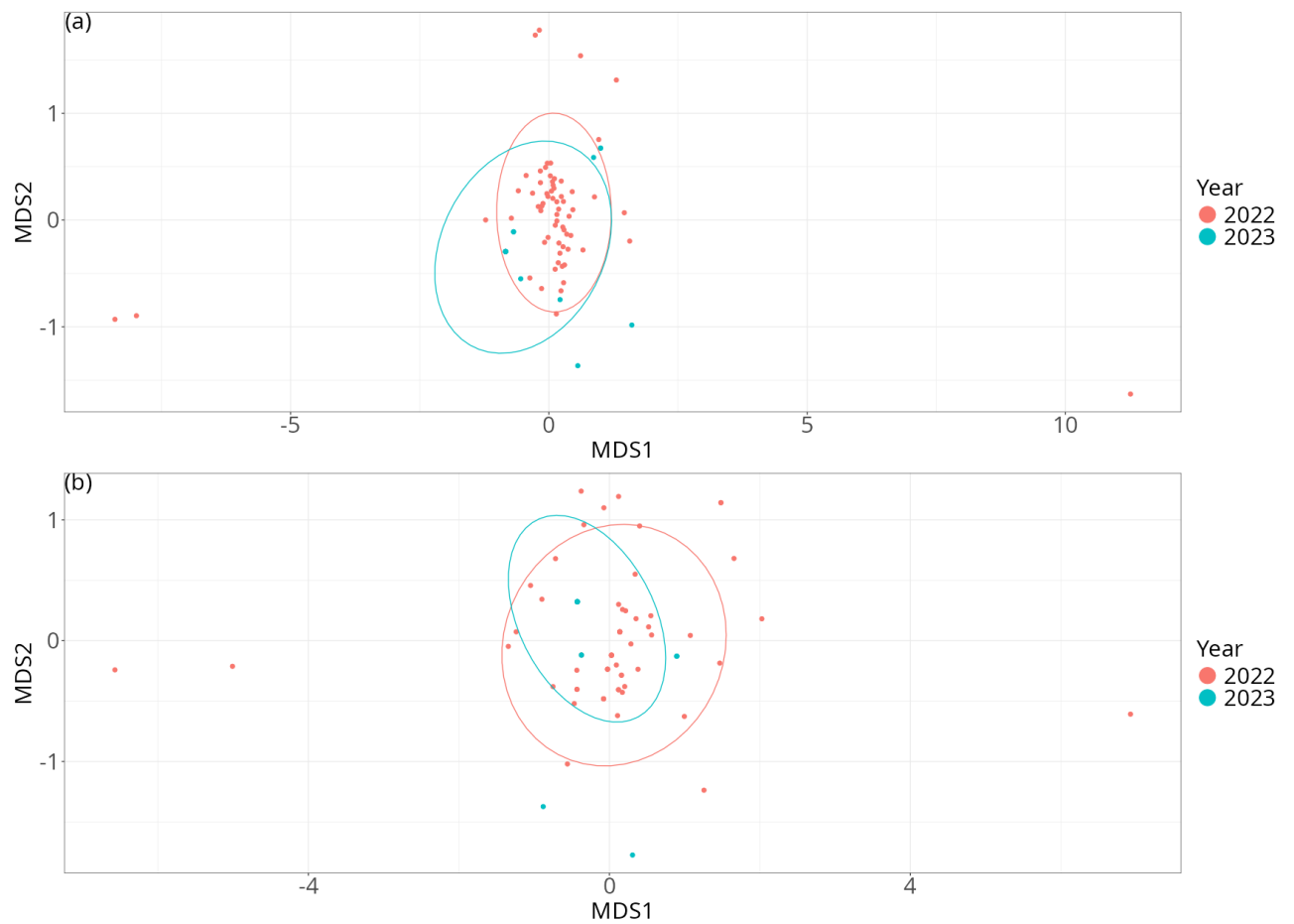


Figure 15: NMDS plot of (a) mean species abundance (stress = 0.1), and (b) species presence (stress = 0.09), compared between the two sampling seasons (October 2022 and March 2023).

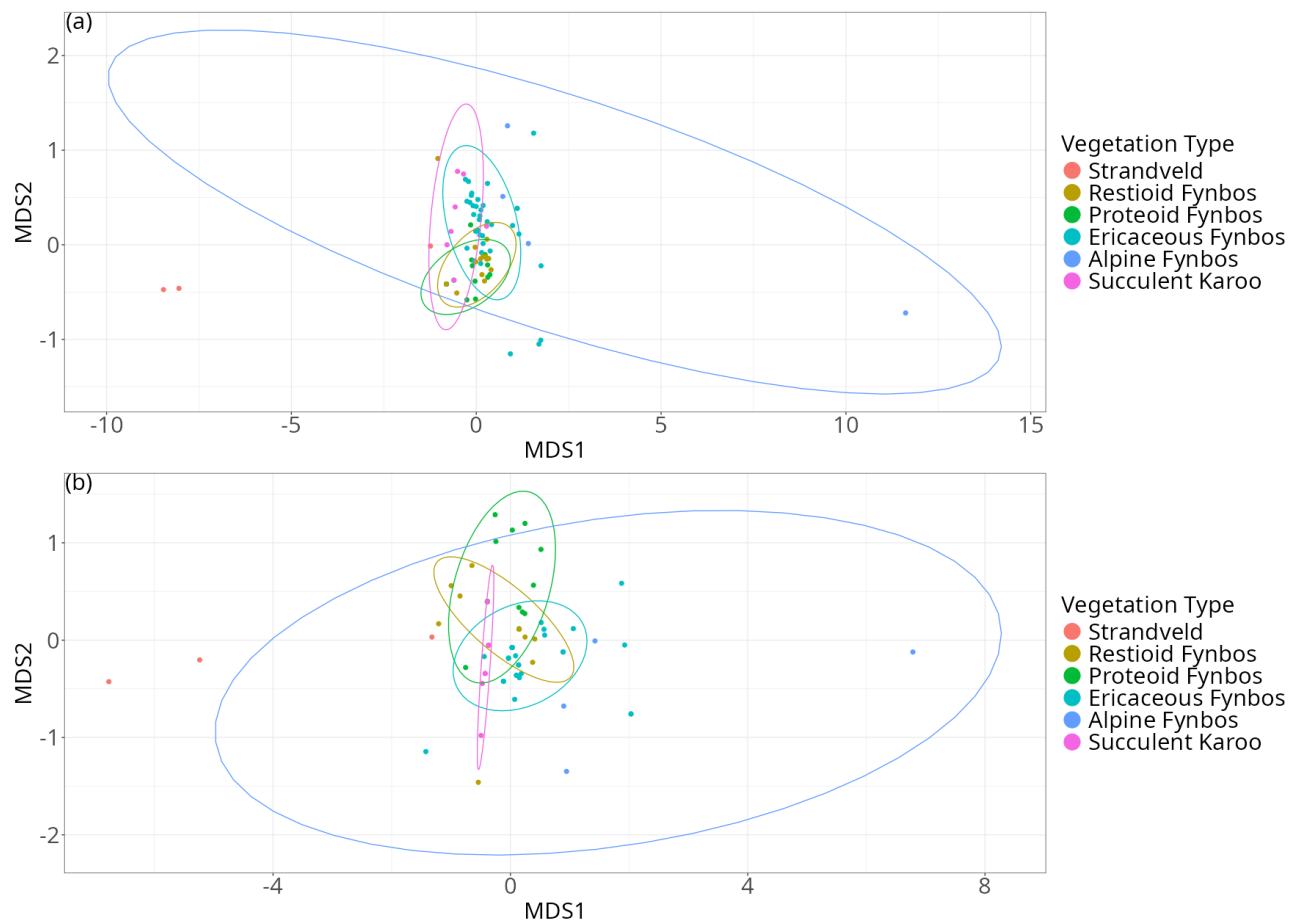


Figure 16: NMDS plot of (a) mean species abundance (stress = 0.1), and (b) species presence (stress = 0.09), compared between the two different vegetation types present across the sites of the transect.

Table 1: Mean species richness and abundance ($\pm$ standard error) of beetles identified from October 2022 across the different vegetation types present within the transect.

Vegetation type	Abundance (mean $\pm$ SE)	Species richness (mean $\pm$ SE)
Alpine Fynbos	8.25 $\pm$ 3.45	2.5 $\pm$ 0.645
Ericaceous Fynbos	32.875 $\pm$ 5.36	3.5 $\pm$ 0.25
Proteoid Fynbos	15.5 $\pm$ 2.85	4.25 $\pm$ 0.629
Restioid Fynbos	19.25 $\pm$ 2.81	3 $\pm$ 0.348
Strandveld	9.25 $\pm$ 3.4	3 $\pm$ 1
Succulent Karoo	23.5 $\pm$ 8.86	3 $\pm$ 0.408

Sample-based species rarefaction curves showed that for most of the sampled sites, the sampled species did not reach the asymptote of the expected number of species present within those sites. Only two sites (900m (W), 1100m (E)) seemed to reach saturation for sampled species (Figure S7).

Table 2: Mean species richness and abundance ($\pm$ standard error) of the beetles identified from the October 2022 sample across the elevational gradient.

Elevation (m a.s.l.)	Abundance (mean $\pm$ SE)	Species richness (mean $\pm$ SE)
Western slope		
0	9.25 $\pm$ 3.4	3 $\pm$ 1
200	17.5 $\pm$ 5.98	4 $\pm$ 0.707
300	24.25 $\pm$ 3.3	6.5 $\pm$ 0.645
500	17 $\pm$ 3.39	3.75 $\pm$ 0.479
700	5.25 $\pm$ 2.32	2.5 $\pm$ 0.957
900	24.25 $\pm$ 4.77	2.75 $\pm$ 0.479
1100	93.5 $\pm$ 20.6	5.5 $\pm$ 0.5
1300	21.5 $\pm$ 3.52	3.5 $\pm$ 0.289
1500	24.25 $\pm$ 6.06	3 $\pm$ 0.408
1700	26.5 $\pm$ 4.19	4.5 $\pm$ 0.289
1900	8.25 $\pm$ 3.45	2.5 $\pm$ 0.645
Eastern slope		
1700	37.75 $\pm$ 6.86	3.75 $\pm$ 0.479
1500	29.5 $\pm$ 8.53	3.5 $\pm$ 0.645
1300	16.5 $\pm$ 13.5	1.25 $\pm$ 0.25
1100	13.5 $\pm$ 6.81	3 $\pm$ 0.577

900	16 ± 3.94	2.25 ± 0.25
500	23.5 ± 8.86	3 ± 0.408

Historical beetle comparisons

Three species were used when performing the historical comparisons, being *Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilocornis*. These species are the three most abundant species in the sampling site across both sampling periods, and were fully identified in both samples, meaning that they could be directly compared over time. Comparing the box plots of abundance and altitudinal site separated by year, it can be observed that in the historical abundances seems generally higher than for recent sampling, especially in 2003 (Figure S10). Additionally, the peaks of abundance for *Anthia decemguttata* and *Zophosis gracilocornis* have shifted from being highest at 1300-1700m W to mostly peaking at 1100m W (Figure 17, Figure 19).

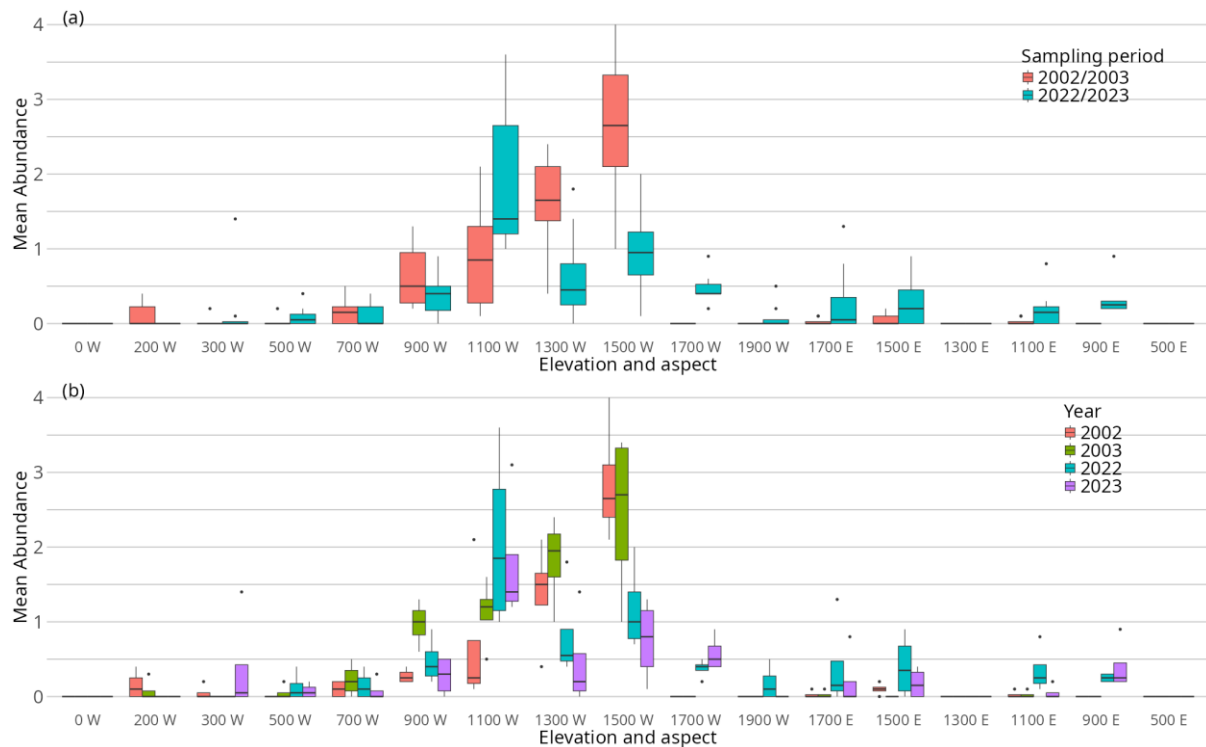


Figure 17: Mean abundance of *Anthia decemguttata* separated by (a) sampling period, and (b) year, across altitudinal gradient.

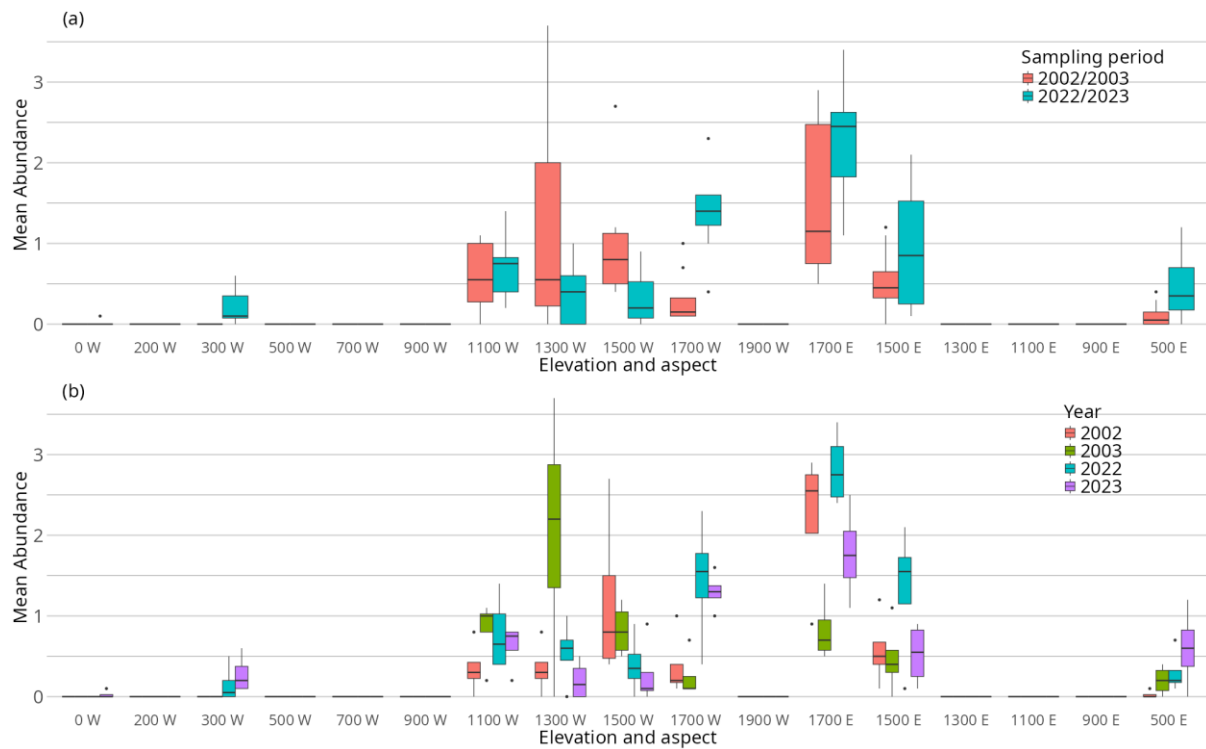


Figure 18: Mean abundance of *Stenocara dentata* separated by (a) sampling period, and (b) year, across altitudinal gradient.

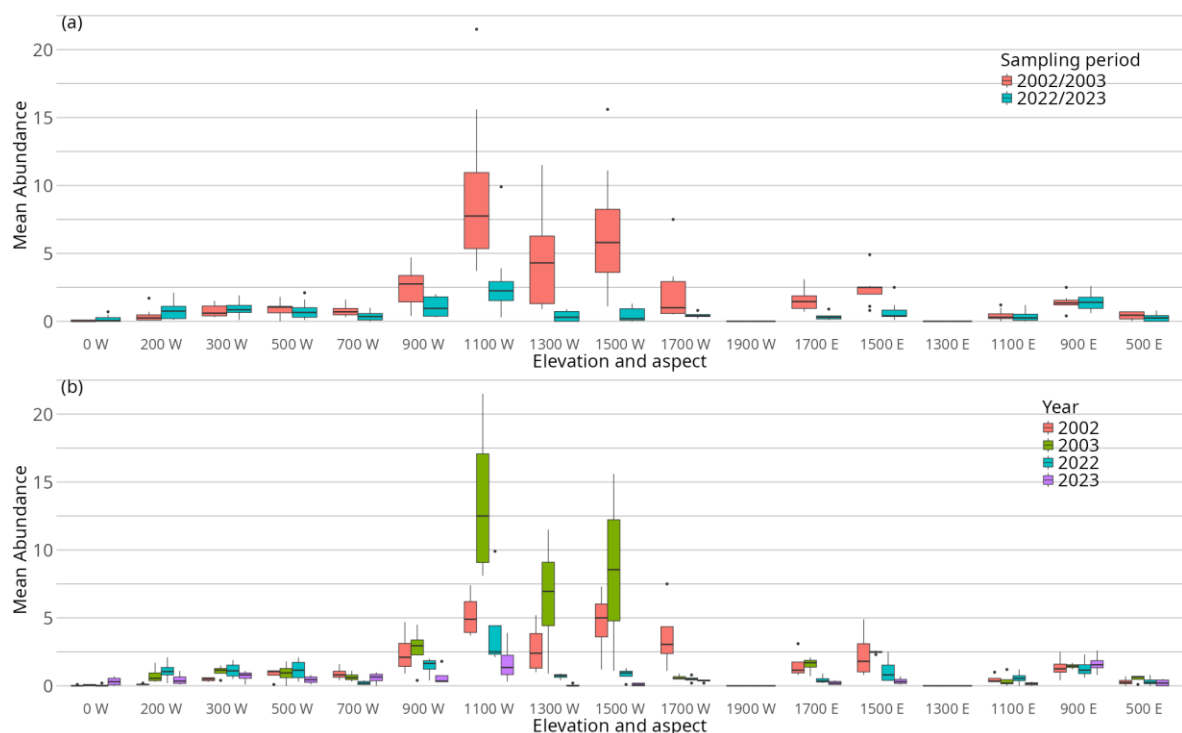


Figure 19: Mean abundance of *Zophosis gracilicornis* separated by (a) sampling period, and (b) year, across altitudinal gradient

The NMDS plots of mean abundance compared between the different sampling years shows that although there is some overlap between the observations from the years of the different sampling periods, there is a clear grouping of the historical samplings when compared to the current ones (Figure S5). This dissimilarity is not as present when comparing the presences of the species between the years, with a more uniform distribution present in the observations (Figure S6). These plots therefore indicate that although the abundance of these three species has changed to some degree over time, the distribution of the populations over the transect has not changed as much.

A heatmap of the Jaccardian similarities between the different altitudinal sites shows which of the sites are the most similar (Figure S8). It shows that different “blocks” of sites are similar to each other both due to matching vegetation types, as well as relative altitudinal placement across the gradient of the transect, even on opposing aspects. The two sites at the peak and at 1300m on the eastern aspect had no recorded observations of any of the key species in every year except for one, and due to multiple rows of zeroes, had to be excluded from the heatmap.

Dendrograms of the different sampling periods show how the different sites are hierarchically grouped with each other when considering beetle abundance (Figure S9). From the clustering of the dendrogram branches it can be observed that in both sampling periods, for the most part the sites are separated into two main groups. The first group consists of the Proteoid and Restioid vegetation sites, and the second group consists of the mid-high altitude Ericaceous sites. The site at 1300m on the eastern slope is different from every single other site, along with the peak of the transect.

The GLMMs were used to quantify the difference in abundance between the years and sampling periods. When all three focus species and sites are considered together, there is a significant 47% decline in abundance ($z = -22.80$, $P < 0.001$, Table S6) between the 2002-2003 and 2022-2023 samples. When considering each of the years individually, there was also a significant difference in abundance between the sites within the sampling periods (Table S2). There was a 43% increase in abundance between October 2002 and 2003 for all species ($z = 10.697$, $P < 0.001$, Table S7). Most of this increase in October 2003 was due to *Zophosis gracilicornis*, experiencing a 54% increase in abundance within that one-year gap ($z = 11.194$, $P < 0.001$, Table

S8). When analysing each individual species by themselves, there were varying observed responses over time. *Anthia decemguttata* did not show significant changes in abundance between sampling periods. *Stenocara dentata* showed a significant 27% increase ($z = 3.713$, $P < 0.001$, Table S9) in abundance over time. *Zophosis* showed a significant 66% decrease ($z = -28.82$, $P < 0.001$, Table S10) in abundance.

The comparisons between species density between the two sampling periods showed that there were minimal differences in density between the three types of Mountain Fynbos and Succulent Karoo, but there were changes in species density observed for the Alpine Fynbos and the Strandveld (Table 3).

Table 3: Mean species density ($\pm$ standard error) of *Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilocornis*, compared between the sampling periods, grouped by vegetation type.

Vegetation type	2002/2003	2022-2023
	(mean $\pm$ SE)	(mean $\pm$ SE)
Strandveld	7 ± 0.567	3 ± 1
Restioid Fynbos	3.67 ± 0.491	3 ± 0.348
Proteoid Fynbos	4.83 ± 0.527	4.25 ± 0.629
Ericaceous Fynbos	3.28 ± 0.197	3.5 ± 0.25
Alpine Fynbos	1.12 ± 0.227	2.5 ± 0.645
Succulent Karoo	2.88 ± 0.227	3 ± 0.408

Explanatory models

For the historical assemblage comparisons, the first two canonical axes together explained 52% of the variance observed when comparing the sample and environment (Axis 1: $F = 229.95$, $P < 0.001$; Axis 2: $F = 169.55$, $P < 0.001$). A total of twelve explanatory factors were selected to significantly explain the beetle abundance variance ($R^2 = 0.63$) (Table S11). Of these variables ten were environmental factors and two were the altitudinal site and the sampling period. The first axis is represented by the environmental gradients of an increase in the mean monthly maximum precipitation, the percentage proportion of calcium in the mineral composition of the soil, wind direction, and a decrease in the vegetation cover and the free calcium ions in the soil. The second axis is represented by an increase in the mean monthly maximum air temperature, the range of the monthly air temperature, mean monthly maximum ground temperature, the percentage proportion of magnesium in the soil, surface incident shortwave radiation, and a decrease in the pH of the soil and the relative humidity (Figure 20).

GLMMs showed that *Zophosis gracilicornis* abundance was affected positively by an increase in the air temperature range, relative atmospheric humidity, percentage of rock, mean maximum monthly precipitation, and changing wind directions. Their abundance had a negative relationship with mean monthly maximum ground and air temperatures (Table S11). *Stenocara dentata* abundance had a positive relationship with the percentage magnesium composition in the soil, calcium content in soil, changing wind directions, an increase in the percentage of rock, ground vegetation cover, solar radiation, and mean monthly maximum precipitation. *Stenocara dentata* had no significant negative relationships with any of the measured environmental variables (Table S12). *Anthia decemguttata* did not change significantly in abundance over time, however their distribution was still altered over time. The environmental variables that affected their distribution over time were percentage magnesium composition in soil, increase in the air temperature range, ground vegetation cover, and mean monthly maximum precipitation. These variables all had a positive

relationship with their distribution across the transect. Conversely, percentage magnesium composition in soil, mean monthly maximum air temperature, and changing wind direction all had a negative effect on their distribution (Table S13).

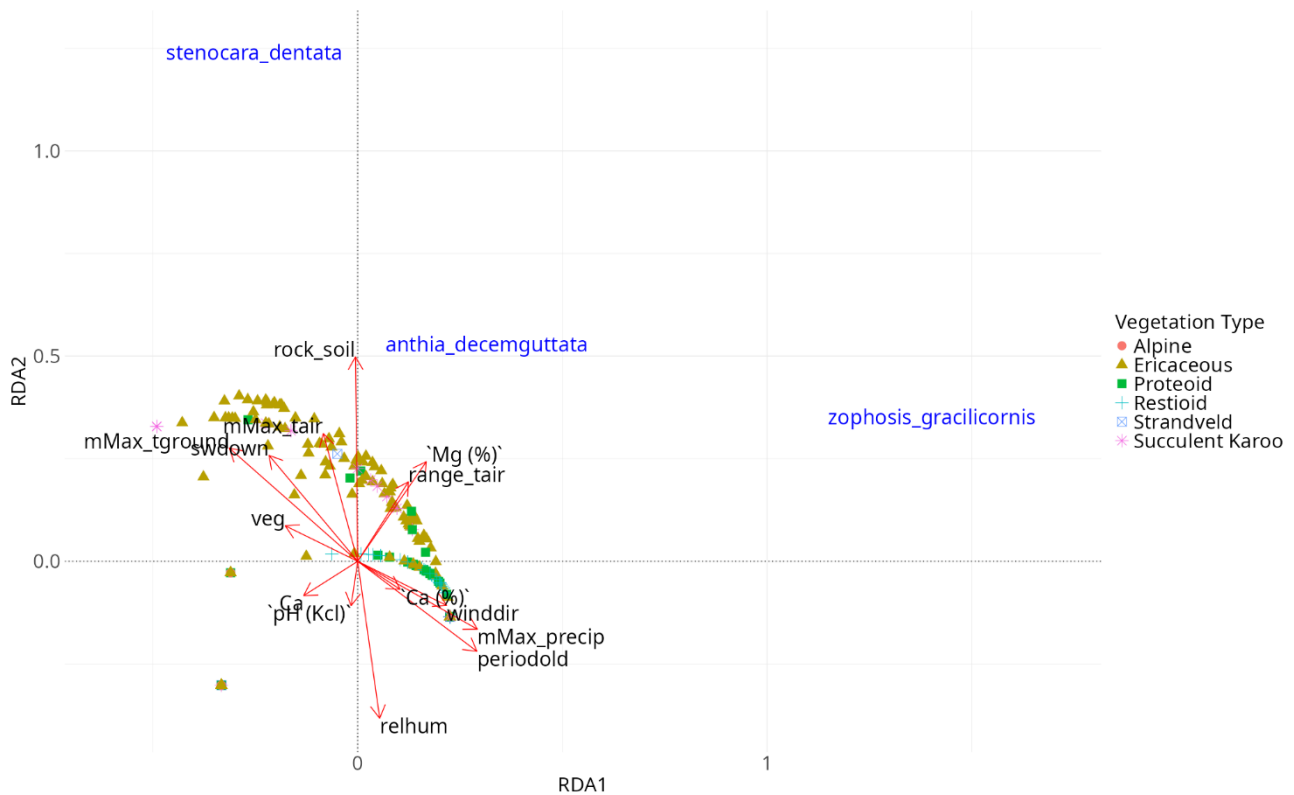


Figure 20: Biplot showing the factors that most significantly explained the variance in the beetle abundance compared between the 2002-2003 sample and 2022-2023 sample, with the relevant vegetation type indicated. (mMax_tair: mean monthly maximum air temperature, mMax_tground: mean monthly maximum ground temperature, swdown: incident shortwave radiation, Mg (%): percentage proportion of magnesium in soil composition, veg: ground vegetation cover, Ca: free calcium ions in soil, Ca (%): percentage proportion of calcium in soil composition, pH (Kcl): pH of the soil, mMax_precip: mean monthly maximum precipitation, relhum: relative humidity).

Due to the extremely low abundances and scattered distribution of species in the March 2023 sample, the models generated had very low explanatory correlation and were excluded from these analyses. When considering the 2022 sample alone, the first two canonical axes together explained 37% of the variance observed when

comparing the sample and environment (Axis 1: $F = 31.0864$, $P < 0.001$; Axis 2: $F = 30.2172$, $P < 0.001$). A total of eight explanatory factors were selected to significantly explain the variance in beetle abundance across all the identified species ($R^2 = 0.59$). The first axis is represented mostly by an increase in the percentage proportion of magnesium in the soil and is represented to a much lesser extent by the ground vegetation cover. The second axis is represented by an increase in the percentage proportion of sodium in the mineral composition of the soil, the pH of the soil, and a decrease in the mean monthly air temperature, monthly range of the ground temperature, and the mean monthly precipitation (Figure 21).

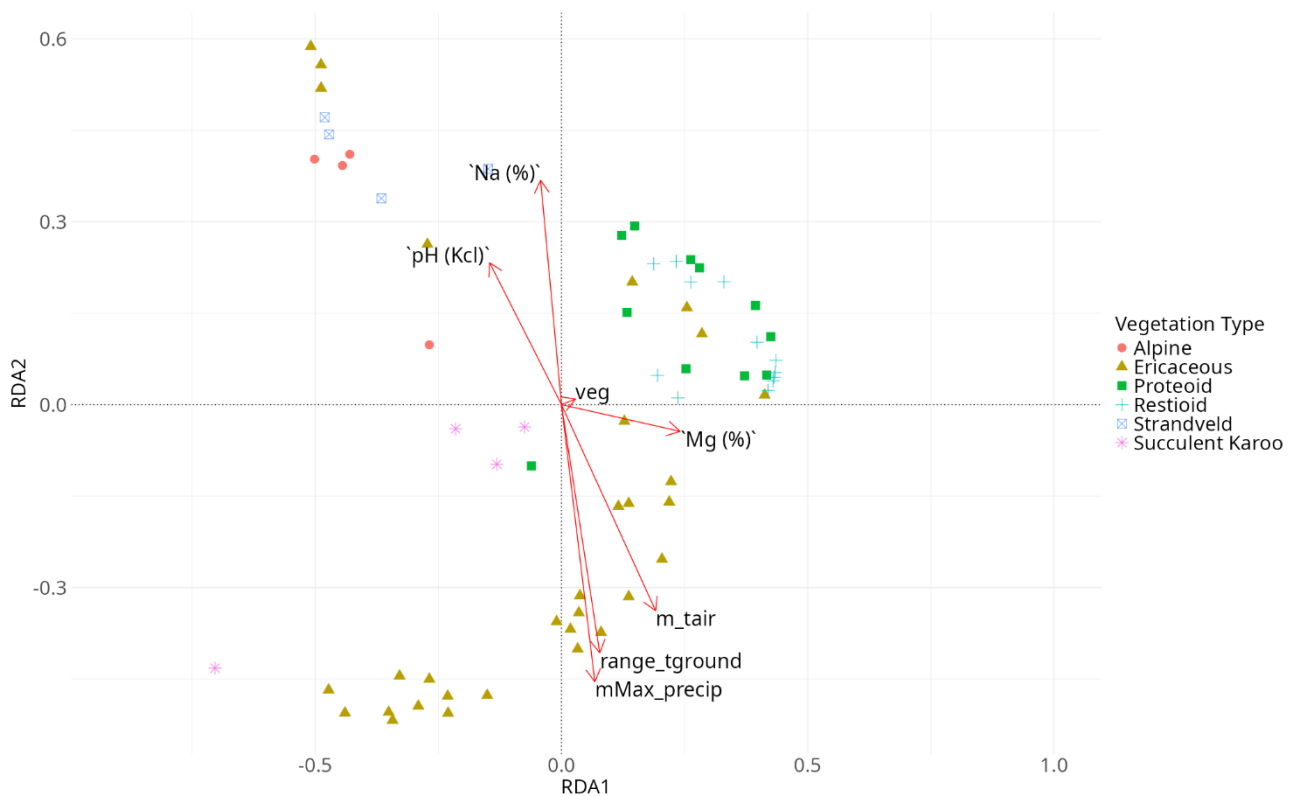


Figure 21: Biplot showing the factors that most significantly explained the variance of all the identified beetle species from October 2022. (Na (%): percentage proportion of sodium in soil composition, pH (Kcl): pH of the soil, veg: ground vegetation cover Mg (%): percentage proportion of magnesium in soil composition, m_tair: mean monthly air temperature, range_tground: monthly range of ground temperature, mMax_precip: mean monthly maximum precipitation).

A representation of the timeline of El Niño and La Niña events over time was created using data provided by the United States National Oceanic and Atmospheric Administration (2026) (Figure 22).

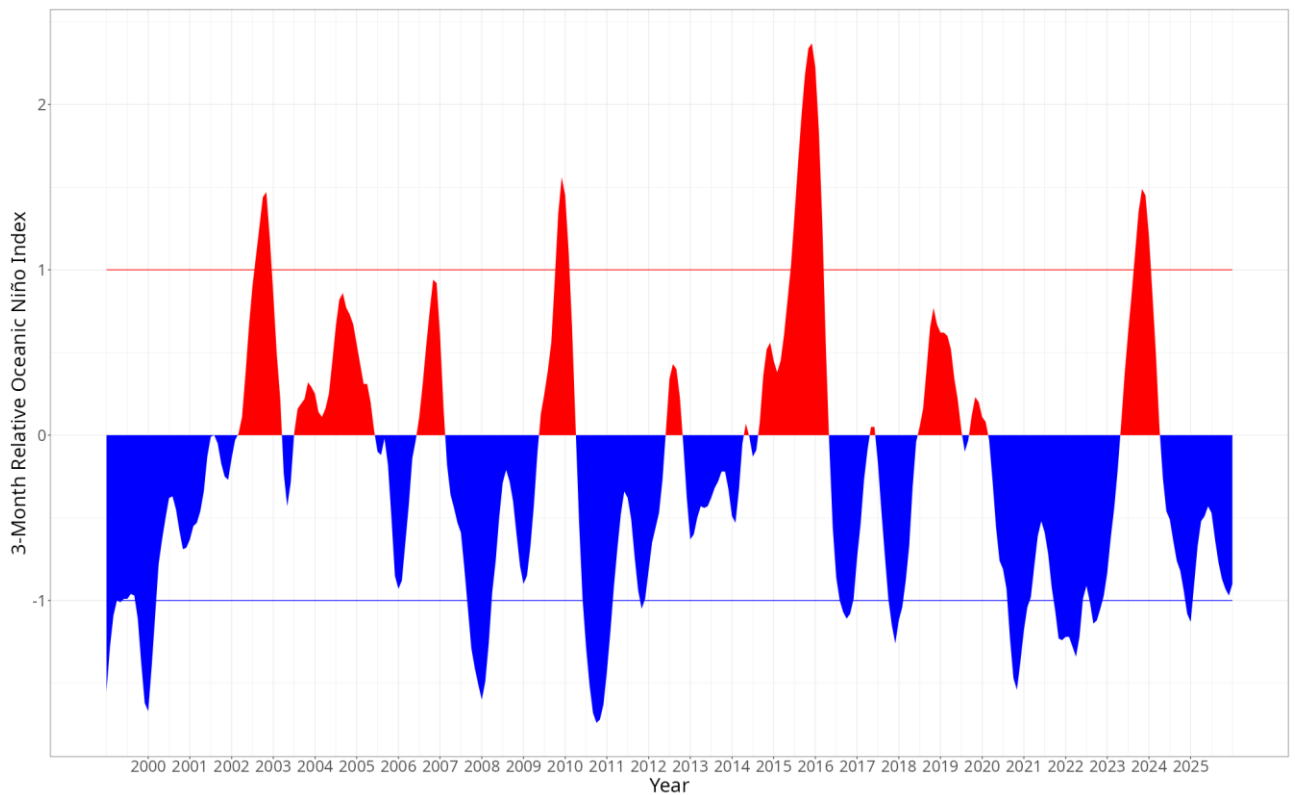


Figure 22: Timeline of El Niño and La Niña events over time. Indices that exceed the red intercept line ($y = 1$) indicate periods of El Niño, and indices that exceed the blue intercept line ($y = -1$) indicate periods of La Niña.

Discussion

Overview of key patterns

Several environmental factors have changed over the 20 years of this study. Ground cover such as live vegetation cover and plant litter have increased, while bare ground and rock cover has decreased. An increasing trend for both ambient air temperatures was observed, while ground-level soil microclimate temperatures, and the short-term and long-term seasonality of both have become more extreme. Precipitation has not exhibited a uniform trend over time. However, the seasonality of rainfall is becoming more extreme; the maximum and minimum ends of the range in precipitation are increasing. Beetle abundance showed no clear trend, although one site had an extreme high abundance, and two sites had extreme low abundance. The highest levels of abundance were found in Ericaceous Fynbos, and the highest levels of richness were found in Proteoid Fynbos. There was a strong seasonal effect with both species richness and abundance being greatly reduced between the October 2022 and March 2023 samples. Overall, there was a decline in abundance between the historical sampling and the recent sampling. This decline was mostly due to one species, *Zophosis gracilicornis*. *Stenocara dentata* showed a significant increase in abundance, and *Anthia decemguttata* showed no significant abundance change. There was a shift in the observed altitudinal distribution for *Anthia decemguttata* and *Zophosis gracilicornis*, with their peaks of occurrence on the western slope of the mountain having shifted downslope. The RDA and GLMM models showed that beetle abundances were related strongly to environmental variables such as temperature extremes, ground compositional cover, and soil variables, but those alone could not explain all the variation reported.

Seasonality vs long-term changes

In this study, beetle abundances decreased by 97% between the October 2022 and March 2023 sampling events, which is comparable to previous studies that also observed similar levels of seasonal beetle population fluctuation (Ndozi, Gohole and Mapaire, 2025). In other Southern Hemisphere studies, the patterns of beetle abundance and prevalence followed a general trend with the difference in the seasonal

samples observed in the Cederberg transect. For example, beetle abundances tend to have multiple seasonal peaks throughout the year with one occurring in the warmer months (in late spring around September/October), being either the maximal peak or one of a few (Pinheiro *et al.*, 2002; Wardhaugh, Stone and Stork, 2018). However, when regarding these prior examples, the warm season is also the season with increased rainfall in their respective environments. Therefore, evidence exists that much of the variation in increasing beetle abundance could be attributed to increasing rain in the warmer months along with the respective temperature changes (Wang *et al.*, 2014).

Other possible reasons behind the large decrease in observed abundance and species richness between seasons could be due to many different species exhibiting reduced physical activity during the colder seasons. In other winter rainfall regions, there have been examples of Tenebrionidae that have periods of diapause as adults, such as the *Pimelia bipunctata* (Fattorini, 2023). Carabidae have also been shown to commonly undergo periods of seasonal diapause as adults (Hodek, 2012). These observations can potentially explain the seasonal assemblage changes observed across this study as occurring due to sampling limitations rather than actual changes in the baseline assemblage for living beetles. Pitfall trapping captures active surface-dwelling arthropods, therefore observed results of pitfall traps usually only adequately represent species that are active in the season within which sampling is being carried out (Byeon *et al.*, 2022). Determining whether the seasonal presence or absence of certain species is due to sampling biases or because of generational turnover will require research into the phenology of the relevant species and whether they experience adult diapause or not.

In other studies regarding summer rainfall regions, rainfall is less of a determining factor in beetle assemblage size and structure than previously thought (Grimbacher and Stork, 2009). Other studies from winter rainfall regions, including this study site, did not report any significant effects of precipitation on the beetle assemblages (Botes, McGeoch and Chown, 2007; Ramírez-Hernández, Micó and Galante, 2014). A potential reason for this might be that since these environments are generally arid compared to other regions, thus the species present are less variable to assemblage

changes due to precipitation. Despite this, the present study focusing on three focal beetle species used in the historical comparison had a positive relationship with increasing levels of maximum precipitation. This could be potentially due to the ecology of the beetle species in this study, especially the detritivorous species, being more affected by the availability and distribution of their preferred vegetative habitats.

Vegetation diversity and biomass have been shown to be the primary driver of diversity and community structure in certain insect groups (Gaona *et al.*, 2025). Additionally, changes in distribution of vegetation can affect insects at various levels of ecological specificity. The distribution ranges of specialist species such as *Papilio cressphontes* (Papilionidae) have been observed to expand rapidly as their host-plant has undergone a range expansion (Wilson *et al.*, 2021). In contrast, generalist species such as the leaf beetle *Cerotoma trifurcata* (Chrysomelidae) have been modelled to indicate their ranges can be heavily restricted by the soybean plants that they consume as their preferred host-plants (Berzitis *et al.*, 2014). The prevalence of the soybean plants that they mostly feed on has meant that the mass availability of food allows the species to have a large distribution range that it currently inhabits.

The ecology of the most abundant beetle in this study's transect (*Zophosis gracilicornis*) is relatively unknown, however whether they are generalists or specialists, environmental vegetation changes might explain why they have undergone an intense loss of abundance and altitudinal range shift. Increasing temperatures have also generally indicated an increase in vegetation and an expansion of vegetation into higher altitudes as temperature conditions make it suitable for them (Zu *et al.*, 2021). As seen in this study, vegetation cover has increased in general across all altitudes, meaning that plant species from different elevations could be shifting into altitudes in which they were previously not present. These vegetative altitudinal range shifts come with an associated increase in competition and a restructuring of the vegetative communities in the relevant environments. Even with increased vegetation cover and range for vegetation in general, the specific host-plants for certain species might be under competitive pressure (Harvey and Malcicka, 2015). As *Zophosis gracilicornis* is found mostly in

Ericaceous Fynbos sites, a particular type of vegetation found within those sites could be a major determining factor as to their distribution and ideal habitat.

One particular site (1300m (E)), exhibited a beetle assemblage that did not match the trends observed in the sites with the same vegetation type as it on the Eastern aspect of the mountain, or the trends in diversity and observed species in general. This site was mostly dominated by a single species—*Cicindela lurida*—and a few small abundances of herbivorous beetles. *Cicindela lurida* was also found in high abundances in another site (1900m). This dominance of a single predatory species is not a recent phenomenon. The October 2002-2003 study also showed the same trend. A plausible reason for this site having a unique composition of beetles could include vegetation differences that were not recorded in this study. Another possible explanation for the absence of predatory Carabidae could be their sensitivity to interspecific competition by other predators or prey availability, especially where predators target similar prey (Niemelä, 1993; Lövei and Sunderland, 1996; Gonçalves *et al.*, 2021).

There was a consistent 0.19°C increase in air temperatures across the entire transect. However, the changes observed at microclimate level temperatures were not consistent between the different altitudinal sites. Some of the sites exhibited ground temperature changes that fluctuated in line with the air temperature changes. However, in certain sites (such as the 700m (W) example), there was a steady increase of soil microclimate temperatures. There are many varied factors that could potentially explain this relationship of greatly increased ground temperatures. Past studies have shown that the greatest increases in soil temperature were usually found in dry environments with sparse vegetation (Harte *et al.*, 1995). However, the vegetation found at these sites were not significantly different from any other site. Another possible reason could include the slope of the site, irrespective of the aspect of the mountain that it was situated on (Adams, 2010). Whatever the case may be for why the soil temperatures in certain sites are increasing in this way, it will result in some taxa being able to shift to these sites as the temperatures there become more suitable for them. Other taxa will not be able to inhabit those areas due their or their prey species' critical thermal limits being exceeded (Harvey and Malcicka, 2015). For

ectotherms, an increase in temperature up to a certain point also means an increase in metabolic activity (Riemer *et al.*, 2018). Previous studies have shown that there is an asymmetrical effect of increased metabolic activity on organisms depending on their trophic level. Increased metabolic activity in predators results in a spike of predatory attacks on herbivores and a disproportionately higher number of herbivores dying to predation (Rudolf and Roman, 2018).

The role of climatic variability

The fluctuations observed in the trends for temperature and precipitation from the seasonal decomposition models lined up closely with the timeline of historical ENSO events that have been recorded (Figure 22) (National Oceanic and Atmospheric Administration, 2026). The temperature increases that were recorded from the beginning of the study show a sharp increase in temperature that coincides with the 2002-2003 ENSO event. However, the temperature that was observed from that point afterwards never reduced below the thresholds reached after that period. The historical study (October 2002) occurred just before an El Niño, and there was a great increase in the abundance of beetles in just one year, with the October 2003 sample having more than double the beetles sampled in 2002 (Table S2). The rapid environmental changes caused by the ENSO could be a potential reason such high abundances were observed. A potential reason for this is that insect sampling may be more determined by insect activity than true population abundance, and ENSO events have been shown to affect the phenology timelines of other insect species (Srygley *et al.*, 2010). There was an El Niño event that started in mid-2023 (Hagen and Azevedo, 2024). Although this El Niño was not particularly intense, fully identifying the beetle samples that were collected in October 2023 could have allowed for an informative explanation regarding rapid responses by the beetle assemblages to environmental changes and a potential comparison to the October 2003 sampling event.

The precipitation changes that were observed as a baseline trend showed no particular pattern over time. ENSO events have been correlated with droughts in the Western Cape previously (Dube, Nhamo and Chikodzi, 2022). However, there has been no direct relationship determined between the effects of ENSO and winter

rainfall, even though it reduces summer rainfall across South Africa (Mahlalela, Blamey and Reason, 2019). The Cape Floristic Region is expected to experience more frequent and intense droughts from the effects of climate change as it progresses (Midgley *et al.*, 2003; Hannah *et al.*, 2005), and Southern Africa in general is also expected to become increasingly more arid from climate change in general (Munday *et al.*, 2025). These projected changes compounded with more intense and frequent ENSO events (Wang *et al.*, 2019) could result in extreme levels of drought and precipitation loss in the CFR. Effects of climate change combined with extreme climate phenomena means that the insect decline currently observed in this environment could increase (Sharp *et al.*, 2025) due to all of the direct and indirect effects.

Current limitations and future considerations

In previous studies that examined how the functional groups that are made up of Tenebrionidae and Carabidae interact with their environment in arid landscapes, local vegetation biomass and vegetation species richness were determined to be two important factors that explained the diversity and abundance of the beetles (Liu *et al.*, 2016). These two environmental variables were not measured in this study. Therefore, a great portion of possible variation that could have been explained by those might be unrepresented in the results. Identifying the vegetation present within sampling locations could also provide evidence that the host-plant driven insect distributions that have been described in past studies also apply to this transect (Berzitis *et al.*, 2014; Wilson *et al.*, 2021; Gaona *et al.*, 2025).

The phenology or ecology of *Zophosis gracilicornis* is not well known, despite being the most abundant beetle in this environment. Knowing whether it has a generalist or specialist diet, how and when it reproduces, or its habitat choices, would help to better understand its extreme loss of abundance. Currently, even though RDA shows that it is negatively affected by increasingly extreme maximum air and ground temperatures, along with sun intensity, it is not clear what effect this has on the species itself. The increased temperatures could be resulting in exceeding of its thermal limits or desiccation resistance. Other studies have proposed that certain species may be responsive to external biotic triggers such as vegetation flowering and leaf flushes to

advance through their life stages (Basset, 2003). This suggests that the observed changes in abundance in the beetle sampling could be due to the peak abundance and periods of activity for this species might have shifted temporally due to changing vegetation characteristics from changing environmental factors (Grimbacher and Stork, 2009).

The beetles in the original study were also identified to morphospecies (Botes, McGeoch and Chown, 2007), although could not be compared to the species in this study. This meant that the historical comparisons had to be limited to the three species that were most abundant and also easily identifiable in the current samples. Not being able to compare all species from the recent study to the historical study meant that the historical comparisons were not able to consider diversity and could only compare abundance. Being able to access and identify the historical samples would allow for diversity comparisons to greatly improve the explanatory extent of this study.

There is evidence that there may be a decline in beetle abundance over the past 20 years. However, most of this decline currently seems to be due to one specific prolific species (*Zophosis gracilicornis*) decreasing by 66%. Currently, there cannot be any conclusive statement made regarding whether there is biodiversity decline occurring in conjunction with the abundance decline. A definitive conclusion on whether the beetle populations of the Cederberg are declining, both in terms of abundance and diversity, can be determined if the limitations mentioned are addressed.

Conclusions

From this study, there have been some apparent changes in the Cederberg over the past 20 years. There has been a compositional change in ground cover across the altitudinal gradient, with vegetative factors such as live vegetation cover and plant litter increasing, while bare ground and rock cover has decreased. The baseline temperature is increasing over time across the entire transect, however some sites are increasing at faster rates than others and not fluctuating. Additionally, extreme climatic phenomena such as ENSO have noticeable effects on the baseline temperature fluctuations over time, however spikes in temperature correlated with ENSO events do not always result in the temperature reducing back down to prior

levels. Along with baseline trend changes, the seasonal fluctuations for both temperature and precipitation are becoming more extreme over time.

The seasonal comparison of beetle samples showed greatly reduced diversity and abundance in the March 2023 sample in comparison to the October 2022 sample. The various most abundant species exhibited altitudinal range shifts between seasons. Both upper limits and lower limits contracted between the different species in the March 2023 sample. When compared to historical collections, there was a great decline in abundance in one of the three focus species (*Zophosis gracilicornis*). Along with this decline in abundance the proportionate peaks of abundance had shifted to be further downslope. The changes in beetle abundance and diversity in both the seasonal and historical comparisons were attributed to a number of the measured environmental variables, with air temperature, ground temperature, and soil contents being consistent significant predictors for both measures of population status.

This is one of the few long-term beetle datasets from South Africa, addressing a critical knowledge gap in insect decline literature. The observations reported from analysing these samples has revealed that some extent of insect decline and environmental change has been occurring in a pristine biodiversity hotspot. These results, and the limitations of the study, should be viewed as a signal to place greater importance on further monitoring in this environment. Including families such as Scarabaeidae, which have been shown to be important pollinators in the CFR (van Kleunen *et al.*, 2007), and developing knowledge of species-specific vegetation interactions, can allow for a more rigorous quantification of the extent and intensity of insect decline in this region. Filling in the “gaps” of variation explained by biotic and abiotic environmental factors can allow for the trends seen in this transect to be extrapolable to other similar environments and aid in future conservation planning.

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Chapter 3: General conclusions

The main aims of this study were to determine whether the beetle assemblage in the Cederberg had changed over time when compared to a historical study (Botes, McGeoch and Chown, 2007). To achieve this aim, various environmental factors present in the area were measured over time, and the insect sampling that was conducted in the historical study was repeated with the exact same configuration and spatial set-up (Botes *et al.*, 2006). From the results, it was discovered that various environmental changes had occurred over the two decades. The composition of the ground cover present within this environment exhibited an increase in the live vegetation cover and plant litter present, and a decrease in bare ground and rocks. Both the ambient air temperatures and the ground-level soil temperatures were gradually increasing, and their seasonality and cycling between extremes were becoming more frequent and more intense. The effects of climatic phenomena such as the El Niño–Southern Oscillation (ENSO) interact with these factors by intensifying hot periods and directly affecting the precipitation present within this environment. Increases in vegetation cover have been linked to rising temperatures and climate warming (Bao *et al.*, 2021).

The recent sampling period showed a massive decline in the abundance of beetles, largely driven by a decline in *Zophosis gracilicornis*. However, due to only three species being identified to species-level and three families being considered, other taxa that fall outside of the scope of this study could also be declining. The decline of the *Zophosis gracilicornis* as observed in this study may indicate the need to do further sampling in this area. For example, dung beetles (Scarabaeidae) have been shown to be particularly affected by insect decline in past studies, and sampling their populations in the Cederberg may reveal more declining populations (Sánchez-Bayo and Wyckhuys, 2019). Many species from the historical study were not identified to species level and therefore, measures of diversity between the two studies could not be directly compared. This limited the comparisons of these samples to just comparing abundances and not being able to compare species richness over time. Previous long-term studies showed that Carabidae did not reduce in abundance but did significantly

decline in diversity, reiterating the importance of species-level identification for comparison to historical assemblages (Homburg *et al.*, 2019).

A variety of environmental factors were compared to the beetle assemblage data to determine the potential drivers of the calculated changes. However, many of the models were limited in that there was a lot of unattributed variance present within the responses of the beetles. These gaps in explainable variance could be reduced by including environmental factors that were not recorded for this study but have been shown to have significant effects on beetle populations. Specifically, vegetation biomass and species diversity, could be incorporated into future analyses (Liu *et al.*, 2016; Gaona *et al.*, 2025). However, it is also probable that a lot of the variance in the beetle populations over time is due to factors that are difficult to measure and quantify. While likely drivers of insect decline have been proposed and supported with evidence (Cardoso *et al.*, 2020), the importance of each of these drivers within the specific areas being considered in any one scenario is relatively unknown. For conservation strategies and efforts to be implemented to minimise the biodiversity lost, experimental studies are necessary in addition to observational studies (Weisser *et al.*, 2023). Other significant factors that should be considered are the specific ecological traits that the different species exhibit. Beetles that exhibit differing feeding ecologies and trophic levels also have differing levels of peak abundance with regard to seasonality (Grimbacher and Stork, 2009). Factors such as larval ecology, reproduction, hibernation, and seasonal peak activity periods all depend on a number of direct climatic environmental variables, as well as potential chemical triggers and signals from surrounding vegetation (Nishida, 2014).

To determine the status of beetle decline in the Cederberg conclusively and for a variety of different species, further studies are needed. In the case that they are declining, including, and identifying other beetle families to examine and describe the specific biological, ecological, physiological, and behavioural traits of the most abundant and intensely changing species need to be described. Lack of knowledge of plant species diversity data, and specific interactions with beetle species, represents the primary limiting factor for further conclusions to be made. However, due to this

area being a biodiversity hotspot, fully assessing and including those factors might not be entirely possible with the current scale and labour availability of this study.

The initially proposed hypothesis that insects would have an upslope altitudinal range shift as temperatures increased (Wilson *et al.*, 2010; Menéndez *et al.*, 2014) was not met in this study. Most likely explanations for this have previously been indicated to be a specific vegetation interaction and being more affected by the distribution of their host-plants due to the benefits these plants provide them (Hodkinson, 2005; McCain and Garfinkel, 2021). Future research should prioritise, where possible, documenting plant species composition, biomass, and host-plant associations along with beetle sampling to gain a holistic overview of the impacts of long-term climatic variation.

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Appendix

Supplementary Tables

Table S1: Generalised Linear Mixed-Effects model comparing percentage ground cover between the 2002-2003 and 2022-2023 sampling periods. Site used as random effect (DIC: 8817.059).

Parameter	Estimate	Lower 95% Confidence Interval	Upper 95% Confidence Interval	Effective Sample Size	p
(Intercept)	42.454	39.908	44.634	1000	<0.001
periodmodern	-14.728	-18.141	-11.712	1000	<0.001
namelitter	-38.415	-41.725	-34.972	1000	<0.001
namerock	-23.167	-26.038	-19.511	896.1	<0.001
nameveg	-8.251	-11.494	-5.225	1000	<0.001
periodmodern:namelitter	22.141	17.769	26.988	1000	<0.001
periodmodern:namerock	12.717	8.215	17.418	1000	<0.001
periodmodern:nameveg	23.99	18.876	28.166	917.8	<0.001

Table S2: Mean abundance ($\pm$ standard error) of *Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilicornis* combined at each elevation in October across the four years of sampling.

Elevation (m a.s.l.)	2002 (mean $\pm$ SE)	2003 (mean $\pm$ SE)	2022 (mean $\pm$ SE)	2023 (mean $\pm$ SE)
Western slope				
0	0.25 $\pm$ 0.25	0.5 $\pm$ 0.289	0.5 $\pm$ 0.5	3.5 $\pm$ 1.55
200	2.75 $\pm$ 0.854	8.5 $\pm$ 2.99	11 $\pm$ 3.89	4.75 $\pm$ 2.25
300	5.25 $\pm$ 1.11	10.5 $\pm$ 2.33	13 $\pm$ 2.94	13.5 $\pm$ 4.91
500	8.25 $\pm$ 2.43	9.75 $\pm$ 3.33	13 $\pm$ 4.71	5.25 $\pm$ 1.8
700	10 $\pm$ 3.03	8.75 $\pm$ 2.69	3.75 $\pm$ 1.6	6.5 $\pm$ 2.72
900	27.25 $\pm$ 8.74	36.75 $\pm$ 9.94	19 $\pm$ 3.39	9.75 $\pm$ 3.35
1100	62.5 $\pm$ 8.45	156 $\pm$ 32.1	71 $\pm$ 17.2	41.25 $\pm$ 9.6
1300	44.75 $\pm$ 14.4	104.25 $\pm$ 29.6	20.75 $\pm$ 3.92	7 $\pm$ 4.85
1500	86.5 $\pm$ 20.3	117.25 $\pm$ 33.8	24 $\pm$ 6.1	11.25 $\pm$ 4.75
1700	40.5 $\pm$ 13.3	8.75 $\pm$ 1.38	23.25 $\pm$ 4.35	22.25 $\pm$ 0.75
1900	0 $\pm$ 0	0 $\pm$ 0	1.75 $\pm$ 1.18	0 $\pm$ 0
Eastern slope				
1700	38 $\pm$ 2.35	24 $\pm$ 4.06	36.75 $\pm$ 6.54	22 $\pm$ 4.49
1500	30 $\pm$ 8.26	29.5 $\pm$ 1.85	28.5 $\pm$ 8.85	10.5 $\pm$ 1.26

1300	0 ± 0	0 ± 0	0 ± 0	0 ± 0
1100	5.25 ± 1.93	4.25 ± 2.66	9.25 ± 3.97	2 ± 0.913
900	13.5 ± 4.33	14.5 ± 1.04	15.5 ± 3.86	20.25 ± 4.09
500	3.25 ± 1.49	7.25 ± 1.11	6.25 ± 1.49	8.25 ± 2.98

Table S3: Mean abundance ($\pm$ standard error) of *Anthia decemguttata*, at each elevation in October across the four years of sampling.

Elevation (m a.s.l.)	2002 (mean $\pm$ SE)	2003 (mean $\pm$ SE)	2022 (mean $\pm$ SE)	2023 (mean $\pm$ SE)
Western slope				
0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
200	6 $\pm$ 0.957	3 $\pm$ 0.75	0 $\pm$ 0	0 $\pm$ 0
300	2 $\pm$ 0.5	0 $\pm$ 0	0 $\pm$ 0	15 $\pm$ 3.42
500	0 $\pm$ 0	2 $\pm$ 0.5	5 $\pm$ 0.946	3 $\pm$ 0.479
700	4 $\pm$ 0.577	9 $\pm$ 1.11	6 $\pm$ 0.957	3 $\pm$ 0.75
900	11 $\pm$ 0.479	39 $\pm$ 1.49	19 $\pm$ 1.55	11 $\pm$ 1.31
1100	27 $\pm$ 4.77	45 $\pm$ 2.29	83 $\pm$ 6.07	71 $\pm$ 4.46
1300	55 $\pm$ 3.54	73 $\pm$ 3.01	33 $\pm$ 3.28	18 $\pm$ 3.23
1500	114 $\pm$ 4.09	98 $\pm$ 5.66	47 $\pm$ 2.95	30 $\pm$ 2.75
1700	0 $\pm$ 0	0 $\pm$ 0	15 $\pm$ 0.629	23 $\pm$ 1.18
1900	0 $\pm$ 0	0 $\pm$ 0	7 $\pm$ 1.18	0 $\pm$ 0
Eastern slope				
1700	1 $\pm$ 0.25	1 $\pm$ 0.25	16 $\pm$ 3.03	8 $\pm$ 2

1500	4 ± 0.408	0 ± 0	16 ± 2.12	7 ± 1.03
1300	0 ± 0	0 ± 0	0 ± 0	0 ± 0
1100	1 ± 0.25	1 ± 0.25	14 ± 1.55	2 ± 0.5
900	0 ± 0	0 ± 0	10 ± 0.289	16 ± 1.68
500	0 ± 0	0 ± 0	0 ± 0	0 ± 0

Table S4: Mean abundance ($\pm$ standard error) of *Stenocara dentata*, at each elevation in October across the four years of sampling.

Elevation (m a.s.l.)	2002 (mean $\pm$ SE)	2003 (mean $\pm$ SE)	2022 (mean $\pm$ SE)	2023 (mean $\pm$ SE)
Western slope				
0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	1 $\pm$ 0.25
200	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
300	0 $\pm$ 0	0 $\pm$ 0	6 $\pm$ 1.19	11 $\pm$ 1.18
500	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
700	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
900	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
1100	14 $\pm$ 1.66	33 $\pm$ 2.1	31 $\pm$ 2.39	25 $\pm$ 1.44
1300	14 $\pm$ 1.66	81 $\pm$ 7.79	22 $\pm$ 2.06	8 $\pm$ 1.22
1500	47 $\pm$ 5.31	33 $\pm$ 1.65	16 $\pm$ 1.87	11 $\pm$ 2.1
1700	15 $\pm$ 2.1	10 $\pm$ 1.5	58 $\pm$ 3.93	52 $\pm$ 1.22
1900	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
Eastern slope				
1700	89 $\pm$ 4.53	33 $\pm$ 2.02	113 $\pm$ 2.32	71 $\pm$ 2.93

1500	23 ± 2.29	19 ± 2.29	53 ± 4.29	21 ± 1.93
1300	0 ± 0	0 ± 0	0 ± 0	0 ± 0
1100	0 ± 0	0 ± 0	0 ± 0	0 ± 0
900	0 ± 0	0 ± 0	0 ± 0	0 ± 0
500	1 ± 0.25	8 ± 0.913	12 ± 1.35	24 ± 2.48

Table S5: Mean abundance ($\pm$ standard error) of *Zophosis gracilicornis*, at each elevation in October across the four years of samplings.

Elevation (m a.s.l.)	2002 (mean $\pm$ SE)	2003 (mean $\pm$ SE)	2022 (mean $\pm$ SE)	2023 (mean $\pm$ SE)
Western slope				
0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	1 $\pm$ 0.25
200	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
300	0 $\pm$ 0	0 $\pm$ 0	6 $\pm$ 1.19	11 $\pm$ 1.18
500	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
700	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
900	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
1100	14 $\pm$ 1.66	33 $\pm$ 2.1	31 $\pm$ 2.39	25 $\pm$ 1.44
1300	14 $\pm$ 1.66	81 $\pm$ 7.79	22 $\pm$ 2.06	8 $\pm$ 1.22
1500	47 $\pm$ 5.31	33 $\pm$ 1.65	16 $\pm$ 1.87	11 $\pm$ 2.1
1700	15 $\pm$ 2.1	10 $\pm$ 1.5	58 $\pm$ 3.93	52 $\pm$ 1.22
1900	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
Eastern slope				
1700	62 $\pm$ 5.27	62 $\pm$ 3.01	18 $\pm$ 1.55	9 $\pm$ 0.75

1500	93 ± 9.35	99 ± 0.629	45 ± 4.96	14 ± 1.32
1300	0 ± 0	0 ± 0	0 ± 0	0 ± 0
1100	20 ± 1.68	16 ± 2.71	23 ± 2.46	6 ± 0.645
900	54 ± 4.33	58 ± 1.04	52 ± 3.63	65 ± 3.71
500	12 ± 1.47	21 ± 1.44	13 ± 1.7	9 ± 1.31

Table S6: Generalised Linear Mixed-Effects model comparing abundance for *Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilicornis* between the sampling periods of 2002-2003 and 2022-2023 (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects				
Groups	Number of obs	Variance	Std.Dev.	
site:replicate	272	2.888	1.7	
Groups:	68			
site:replicate				
Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.40281	0.21101	11.39	<2e-16
periodmodern	-0.63808	0.02798	-22.8	<2e-16

Table S7: Generalised Linear Mixed-Effects model comparing abundance for *Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilicornis* between the four years in which sampling occurred in October (2002, 2003, 2022, 2023) (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects			
Groups	Number of obs	Variance	Std.Dev.
site:replicate	272	2.888	1.7

Groups: 68

site:replicate

Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.20812	0.21193	10.419	2.00E-16
year2003	0.3576	0.03343	10.697	2.00E-16
year2022	-0.24032	0.03865	-6.218	5.05E-10
year2023	-0.69845	0.0445	-15.697	2.00E-16

Table S8: Generalised Linear Mixed-Effects model comparing abundance for *Zophosis gracilicornis* between the four years in which sampling occurred in October (2002, 2003, 2022, 2023) (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects				
Groups	Number of obs	Variance	Std.Dev.	
site:replicate	272	2.741	1.656	
Groups:	68			
site:replicate				
Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.89362	0.20874	9.072	<2e-16
year2003	0.43456	0.03882	11.194	<2e-16
year2022	-0.57988	0.05048	-11.487	<2e-16
year2023	-1.21697	0.06328	-19.232	<2e-16

Table S9: Generalised Linear Mixed-Effects model comparing abundance for *Stenocara dentata* between the sampling periods of 2002-2003 and 2022-2023 (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects				
Groups	Number of obs	Variance	Std.Dev.	
site:replicate	272	12.56	3.544	
Groups:	68			
site:replicate				
Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.05907	0.65839	-3.127	0.001763
periodmodern	0.24201	0.06518	3.713	0.000205

Table S10: Generalised Linear Mixed-Effects model comparing abundance for *Zophosis gracilicornis* between the sampling periods of 2002-2003 and 2022-2023 (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects				
Groups	Number of obs	Variance	Std.Dev.	
site:replicate	272	2.741	1.656	
Groups:	68			
site:replicate				
Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.13432	0.2074	10.29	<2e-16
periodmodern	-1.08923	0.03779	-28.82	<2e-16

Table S11: Generalised Linear Mixed-Effects model comparing abundance for *Zophosis gracilicornis* and the effects of environmental variables over time (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects				
Groups	Number of obs	Variance	Std.Dev.	
site:replicate	272	2.185	1.478	
Groups:	68			
site:replicate				
Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.68E+01	2.71E+01	-0.991	3.22E-01
periodmodern	-9.41E+01	7.58E+01	-1.241	2.15E-01
`Mg(%)`	9.46E-03	1.86E-02	0.509	0.61045
Ca	1.57E-01	2.01E-01	0.782	0.43415
`Ca(%)`	-1.28E-02	1.11E-02	-1.161	0.24557
mMax_tair	-1.12E+00	3.75E-01	-2.986	0.00282
winddir	1.35E-01	2.91E-02	4.626	3.72E-06
range_tair	1.31E+00	1.51E-01	8.678	<2.00E-16
`pH(Kcl)`	7.63E-02	1.88E-01	0.405	0.68535

relhum	2.11E-01	7.20E-02	2.934	0.00335
swdown	-8.44E-02	5.34E-02	-1.581	0.11382
mMax_tground	-6.79E-02	3.06E-02	-2.215	0.02675
rock_soil	3.05E-02	1.21E-02	2.525	0.01155
veg	2.53E-04	3.05E-03	0.083	0.93377
mMax_precip	1.97E+01	3.87E+00	5.097	3.45E-07
periodmodern:`Mg(%)`	-3.27E-02	1.56E-02	-2.096	0.03604
periodmodern:Ca	-1.56E+00	2.57E-01	-6.071	1.27E-09
periodmodern:`Ca(%)`	6.20E-02	1.24E-02	5.018	5.22E-07
periodmodern:mMax_tair	1.20E+00	4.24E-01	2.827	0.0047
periodmodern:winddir	-6.32E-02	2.49E-02	-2.536	0.01122
periodmodern:range_tair	-1.07E+00	1.86E-01	-5.769	8.00E-09
periodmodern:`pH(Kcl)`	4.77E-01	2.32E-01	2.056	0.03975
periodmodern:relhum	-1.42E-01	1.23E-01	-1.15	0.25003
periodmodern:swdown	4.26E-01	2.22E-01	1.92	0.05484
periodmodern:mMax_tground	5.15E-03	3.11E-02	0.166	0.86846
periodmodern:rock_soil	-1.33E-01	2.42E-02	-5.505	3.69E-08

periodmodern:veg	-2.72E-02	6.01E-03	-4.531	5.87E-06
periodmodern:mMax_precip	-1.56E+01	1.29E+01	-1.211	0.22587

Table S12: Generalised Linear Mixed-Effects model comparing abundance for *Stenocara dentata* and the effects of environmental variables over time (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects				
Groups	Number of obs	Variance	Std.Dev.	
site:replicate	272	7.408	2.722	
Groups: site:replicate	68			
Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.32E+02	6.07E+01	-2.182	2.91E-02
periodmodern	1.60E+02	3.65E+01	4.379	1.19E-05
`Mg(%)`	1.13E-01	4.11E-02	2.741	0.00613
Ca	1.42E+00	4.76E-01	2.986	0.002825
`Ca(%)`	-6.01E-02	3.72E-02	-1.616	0.106195
mMax_tair	-1.97E-01	9.79E-01	-0.201	0.840626
winddir	1.71E-01	8.22E-02	2.079	3.77E-02
range_tair	-4.67E-01	6.59E-01	-0.708	0.478654
`pH(Kcl)`	-7.79E-02	6.59E-01	-0.118	0.905971

relhum	-4.77E-01	2.88E-01	-1.656	0.097687
swdown	3.96E-01	1.26E-01	3.157	0.001596
rock_soil	1.15E-01	2.20E-02	5.238	1.62E-07
veg	2.74E-02	8.58E-03	3.197	0.001391
mMax_precip	1.07E+02	1.96E+01	5.438	5.38E-08
periodmodern:`Mg(%)`	-9.56E-02	3.53E-02	-2.71E+00	0.006798
periodmodern:Ca	-1.71E+00	5.19E-01	-3.294	0.000989
periodmodern:`Ca(%)`	1.49E-01	3.78E-02	3.94E+00	8.33E-05
periodmodern:mMax_tair	7.46E-01	9.20E-01	0.81	4.18E-01
periodmodern:winddir	-2.33E-01	5.90E-02	-3.959	7.52E-05
periodmodern:range_tair	5.24E-01	6.99E-01	0.75	0.453221
periodmodern:`pH(Kcl)`	8.04E-01	5.90E-01	1.36E+00	0.173349
periodmodern:relhum	1.53E-01	2.97E-01	0.514	0.607434
periodmodern:swdown	-4.80E-01	3.07E-02	-15.63	<2.00E-16
periodmodern:rock_soil	1.18E-01	4.85E-02	2.425	0.015294
periodmodern:veg	-2.19E-02	1.13E-02	-1.939	0.052466
periodmodern:mMax_precip	-4.89E+01	4.34E+01	-1.127	2.60E-01

Table S13: Generalised Linear Mixed-Effects model comparing abundance for *Anthia decemguttata* and the effects of environmental variables over time (Poisson error distribution, log link function). Sites and replicates used as random effects.

Random effects				
Groups	Number of obs	Variance	Std.Dev.	
site:replicate	272	4.513	2.124	
Groups:	68			
site:replicate				
Fixed effects	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.04E+02	6.34E+01	1.632	1.03E-01
periodmodern	-3.81E+02	2.94E+01	-12.959	<2.00E-16
`Mg(%)`	1.43E-01	3.49E-02	4.09E+00	4.24E-05
Ca	5.66E-01	3.47E-01	1.632	1.03E-01
`Ca(%)`	-5.33E-02	2.42E-02	-2.203	0.027565
mMax_tair	-3.00E+00	8.01E-01	-3.743	0.000182
winddir	-2.02E-01	8.61E-02	-2.349	0.018837
range_tair	1.32E+00	3.19E-01	4.137	3.51E-05
`pH(Kcl)`	7.71E-02	4.16E-01	1.85E-01	0.852915

relhum	1.97E-01	1.56E-01	1.26	0.207514
swdown	-2.13E-01	1.03E-01	-2.066	0.038784
rock_soil	4.78E-02	2.52E-02	1.899	0.057513
veg	1.66E-02	6.36E-03	2.6	0.009311
mMax_precip	6.89E+01	1.06E+01	6.484	8.92E-11
periodmodern:`Mg(%)`	-1.20E-01	2.67E-02	-4.49	7.13E-06
periodmodern:Ca	-7.91E-01	4.51E-01	-1.755	7.93E-02
periodmodern:`Ca(%)`	1.09E-01	2.32E-02	4.673	2.97E-06
periodmodern:mMax_tair	6.68E+00	9.42E-01	7.09	1.34E-12
periodmodern:winddir	-4.73E-02	6.68E-02	-7.08E-01	4.79E-01
periodmodern:range_tair	-2.18E-01	4.67E-01	-0.467	6.40E-01
periodmodern:`pH(Kcl)`	7.05E-01	5.93E-01	1.19E+00	0.234377
periodmodern:relhum	-5.97E-01	1.81E-01	-3.306	0.000946
periodmodern:swdown	9.52E-01	3.28E-02	2.90E+01	<2.00E-16
periodmodern:rock_soil	-4.44E-02	5.61E-02	-0.791	0.428733
periodmodern:veg	-2.78E-02	1.02E-02	-2.727	0.006388
periodmodern:mMax_precip	-5.41E-02	3.73E+01	-0.001	0.998843

Table S14: total abundance of beetle species identified from Tenebrionidae, Carabidae, and Cicindelidae.

Species	Family	2022	2023
<i>Stenocara dentata</i>	Tenebrionidae	311	24
<i>Zophosis gracilicornis</i>	Tenebrionidae	607	13
<i>Anthia decemguttata</i>	Carabidae	271	4
<i>Cryptochile</i> sp. 1	Carabidae	122	2
<i>Gonogenia</i> sp. 1	Tenebrionidae	23	1
<i>Hypomelus</i> sp. 1	Tenebrionidae	4	0
<i>Stenocara</i> sp. 1	Tenebrionidae	6	0
<i>Oncotus cedrimontis</i>	Tenebrionidae	3	0
<i>Cryptochile assimilis</i>	Tenebrionidae	2	0
<i>Horatoma</i> sp. 1	Tenebrionidae	13	0
<i>Cryptochile setipennis</i>	Tenebrionidae	6	0
<i>Cryptochile</i> sp. 2	Tenebrionidae	9	0
<i>Pachynotelus</i> sp. 1	Tenebrionidae	3	0

<i>Pachynotelus</i> sp. 2	Tenebrionidae	12	0
<i>Moluris globulicollis</i>	Tenebrionidae	16	0
<i>Mariazofia</i> sp. 1	Tenebrionidae	1	0
<i>Ocnodes</i> sp. 1	Tenebrionidae	1	0
<i>Ocnodes</i> sp. 2	Tenebrionidae	2	0
<i>Ocnodes</i> sp. 3	Tenebrionidae	1	0
<i>Ocnodes</i> sp. 4	Tenebrionidae	1	0
<i>Ocnodes</i> sp. 5	Tenebrionidae	1	0
<i>Psammodes longicornis</i>	Tenebrionidae	1	0
<i>Somaticus</i> sp. 1	Tenebrionidae	1	0
<i>Somaticus</i> sp. 2	Tenebrionidae	8	0
<i>Somaticus</i> sp. 3	Tenebrionidae	6	0
<i>Physadesmia</i> sp. 1	Tenebrionidae	1	0
<i>Oxura</i> sp. 1	Tenebrionidae	2	0
<i>Somaticus</i> sp. 4	Tenebrionidae	14	0
<i>Somaticus</i> sp. 5	Tenebrionidae	1	0

<i>Hypomelus</i> sp. 2	Tenebrionidae	1	0
<i>Ocnodes</i> sp. 6	Tenebrionidae	1	0
<i>Broomium</i> sp. 1	Tenebrionidae	8	0
<i>Broomium</i> sp. 2	Tenebrionidae	1	0
<i>Herpiscius spinolae</i>	Tenebrionidae	1	0
<i>Planostibes</i> sp. 1	Tenebrionidae	3	0
<i>Blenosia</i> sp. 1	Tenebrionidae	1	0
<i>Harpalus</i> sp. 1	Carabidae	1	0
<i>Harpalus</i> sp. 2	Carabidae	12	0
<i>Hystrichopus</i> sp. 1	Carabidae	2	0
<i>Scarites rugosus</i>	Carabidae	16	0
<i>Cicindela lurida</i>	Cicindelidae	76	0
<i>Cicindela</i> sp. 1	Cicindelidae	17	0

Supplementary Figures

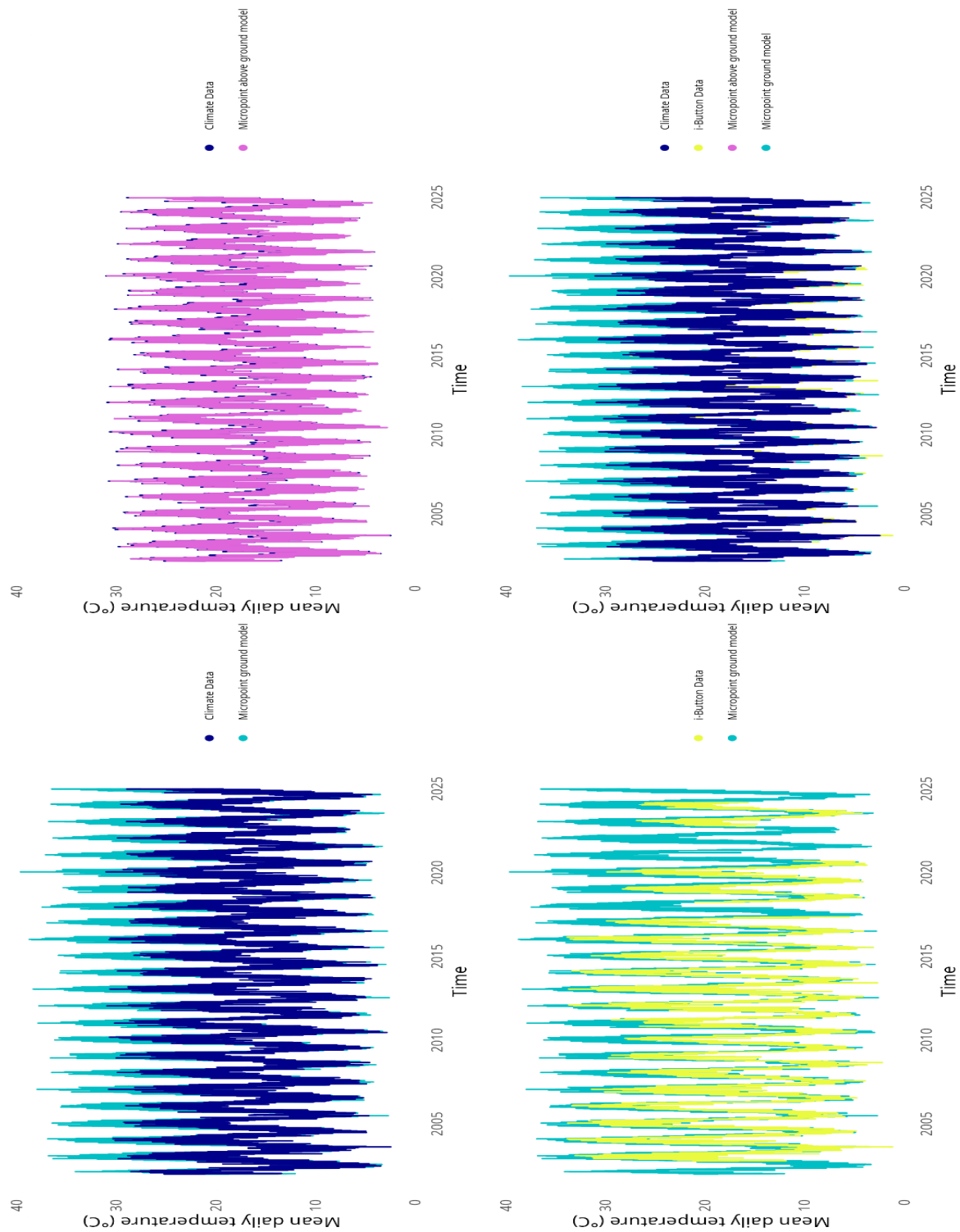


Figure S1: Comparisons between the different sources of temperature data recorded for this study.

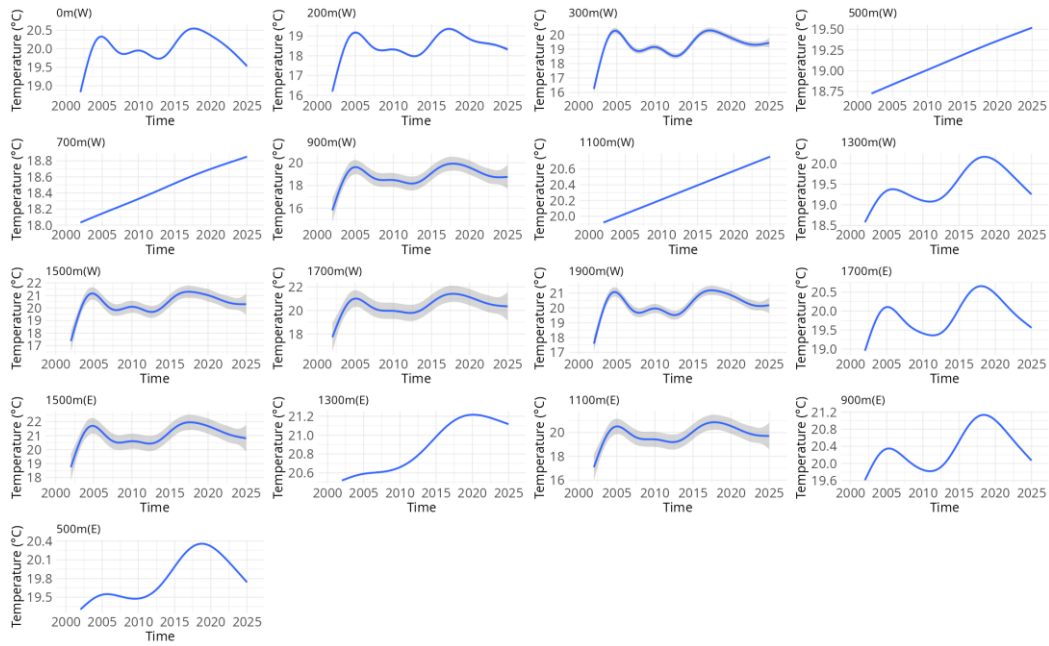


Figure S2: Trend in baseline precipitation for each site across the entire transect.

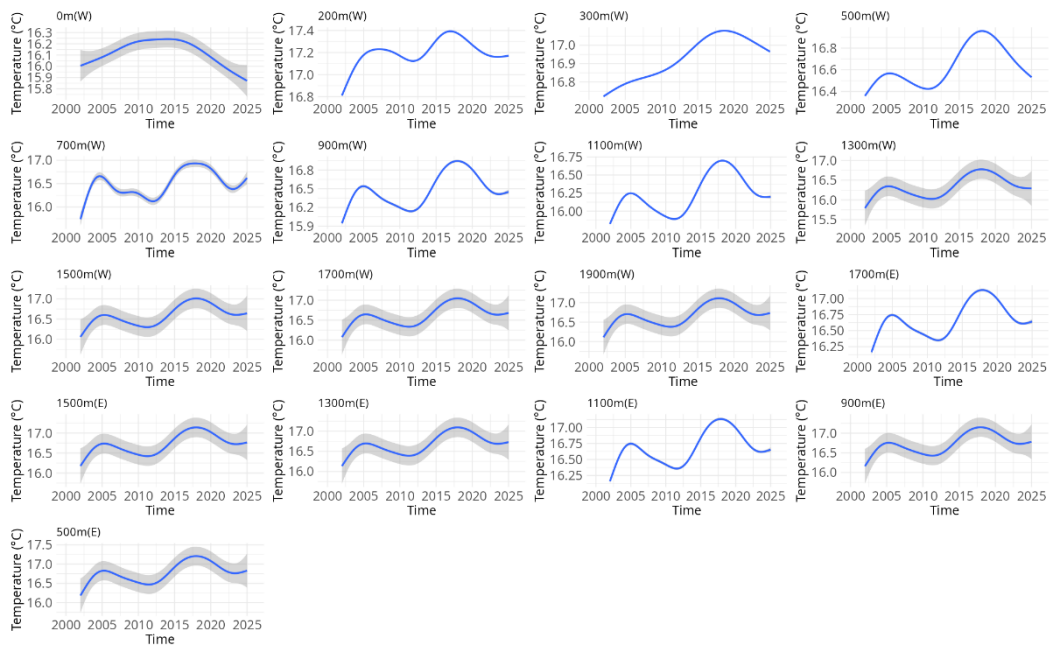


Figure S3: Temperature trend over time for (top) ground temperature, and (bottom) air temperature for each site across the transect.

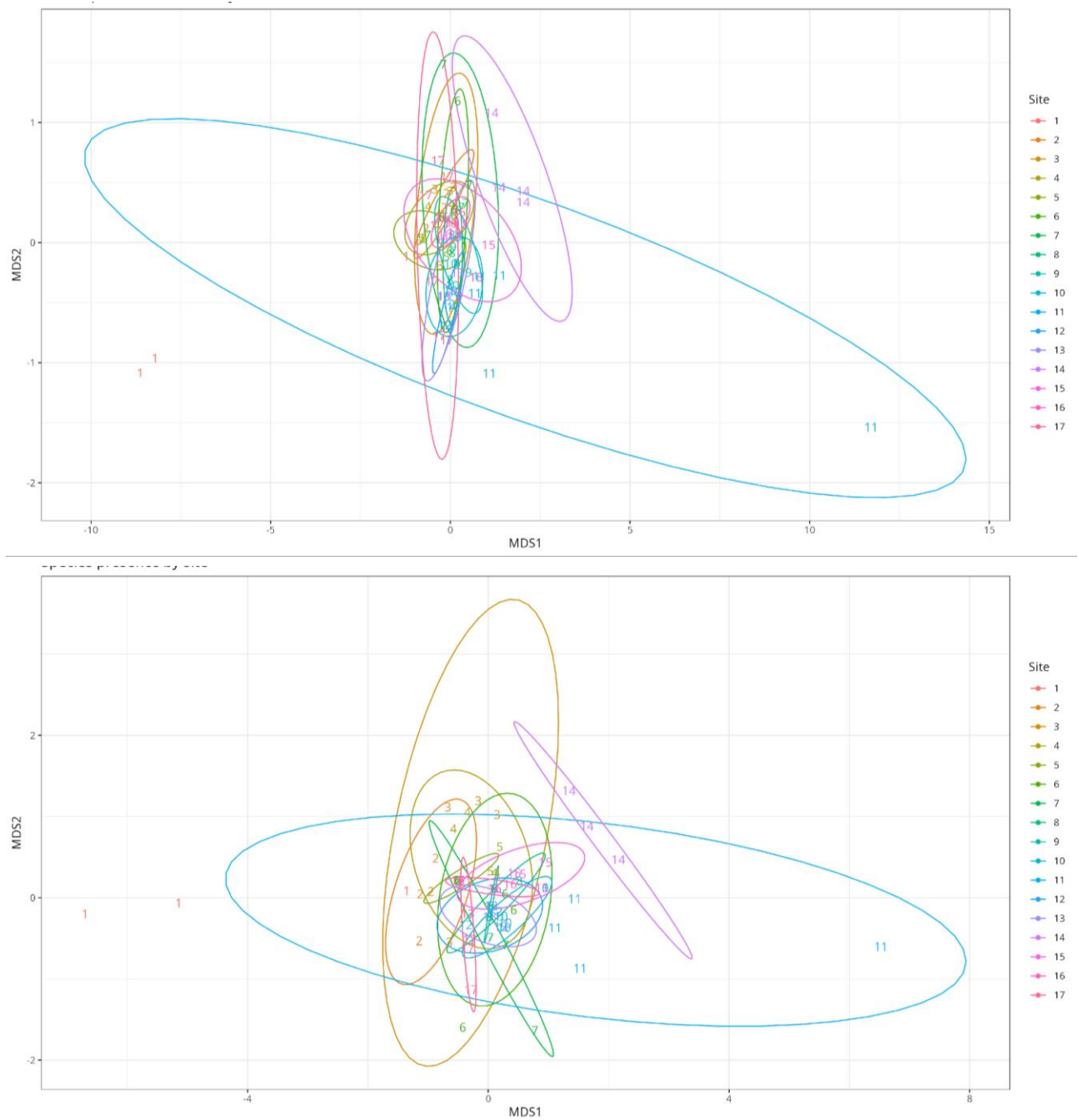


Figure S4: NMDS plot of (a) mean species abundance (stress = 0.1), and (b) species presence (stress = 0.09), compared between the different sites across the altitudinal gradient of the transect.

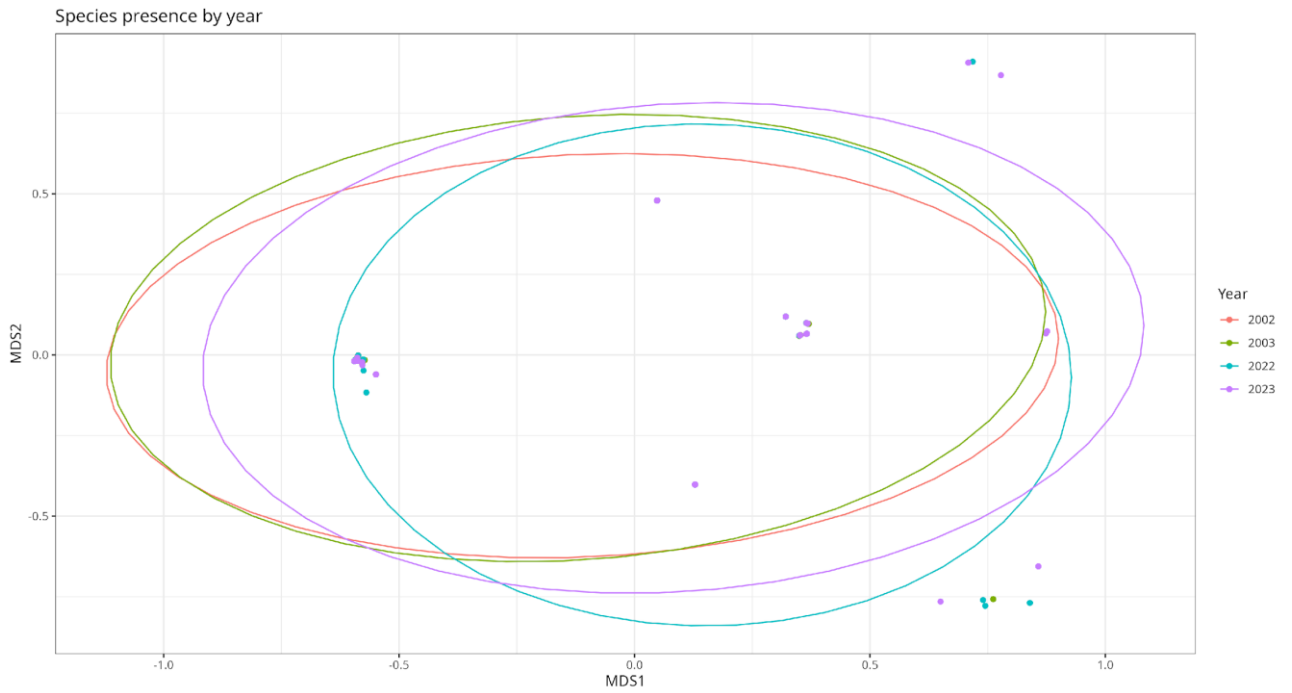


Figure S5: Abundance of the three key species separated by sampling year (stress = 0.16).

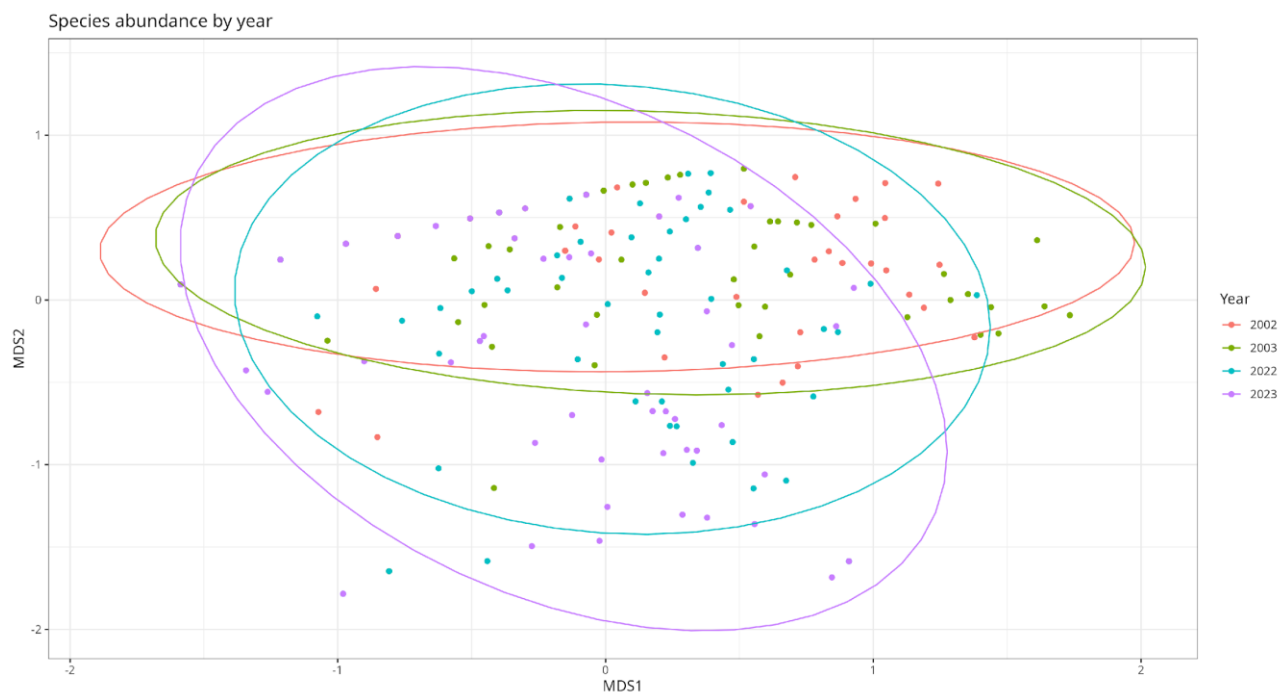


Figure S6: Presence of the three key species separated by sampling year (stress = $1.7e-05$).

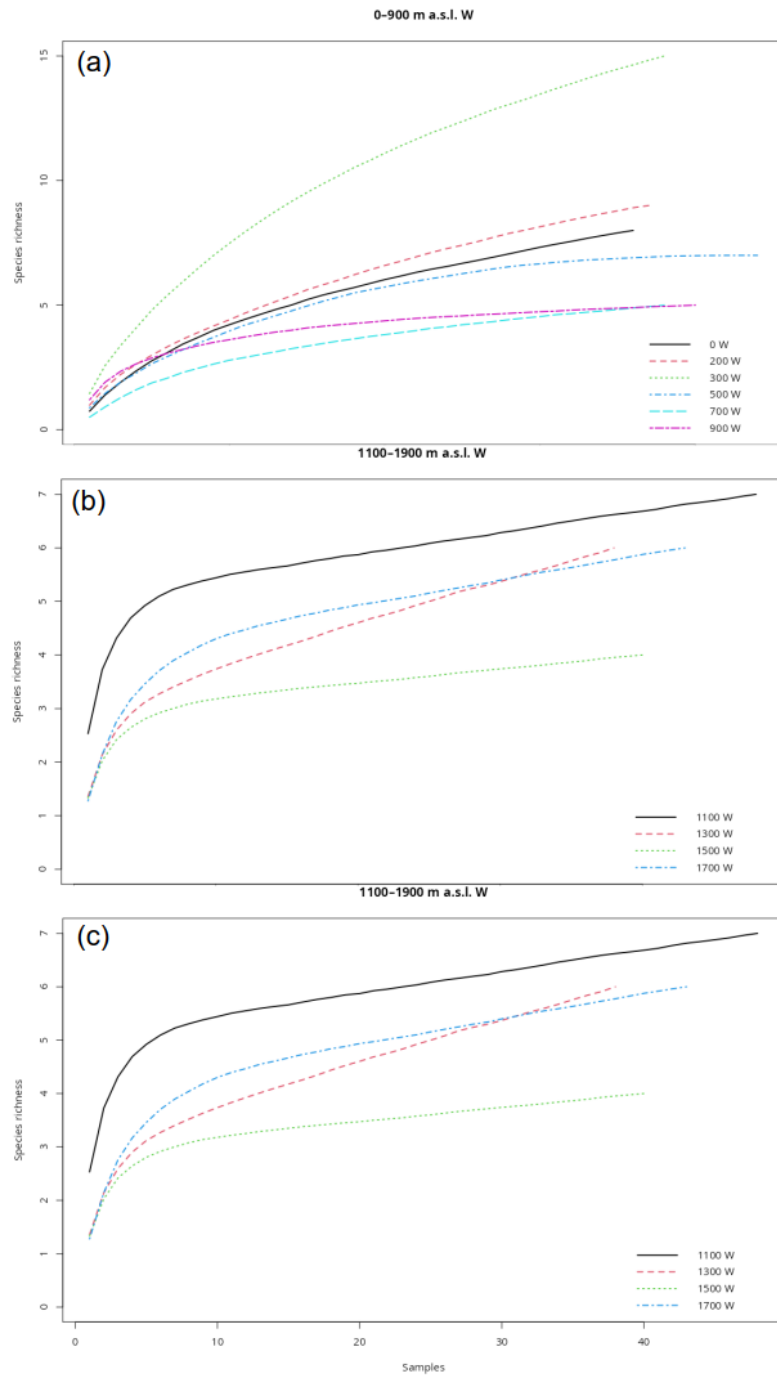


Figure S7 Sample-based rarefaction curves of beetle assemblages at (a) sea level to 900m above sea level on the western slope, (b) 1100–1900m above sea level on the western slope, and (c) 1700–500m above sea level on the eastern slope.

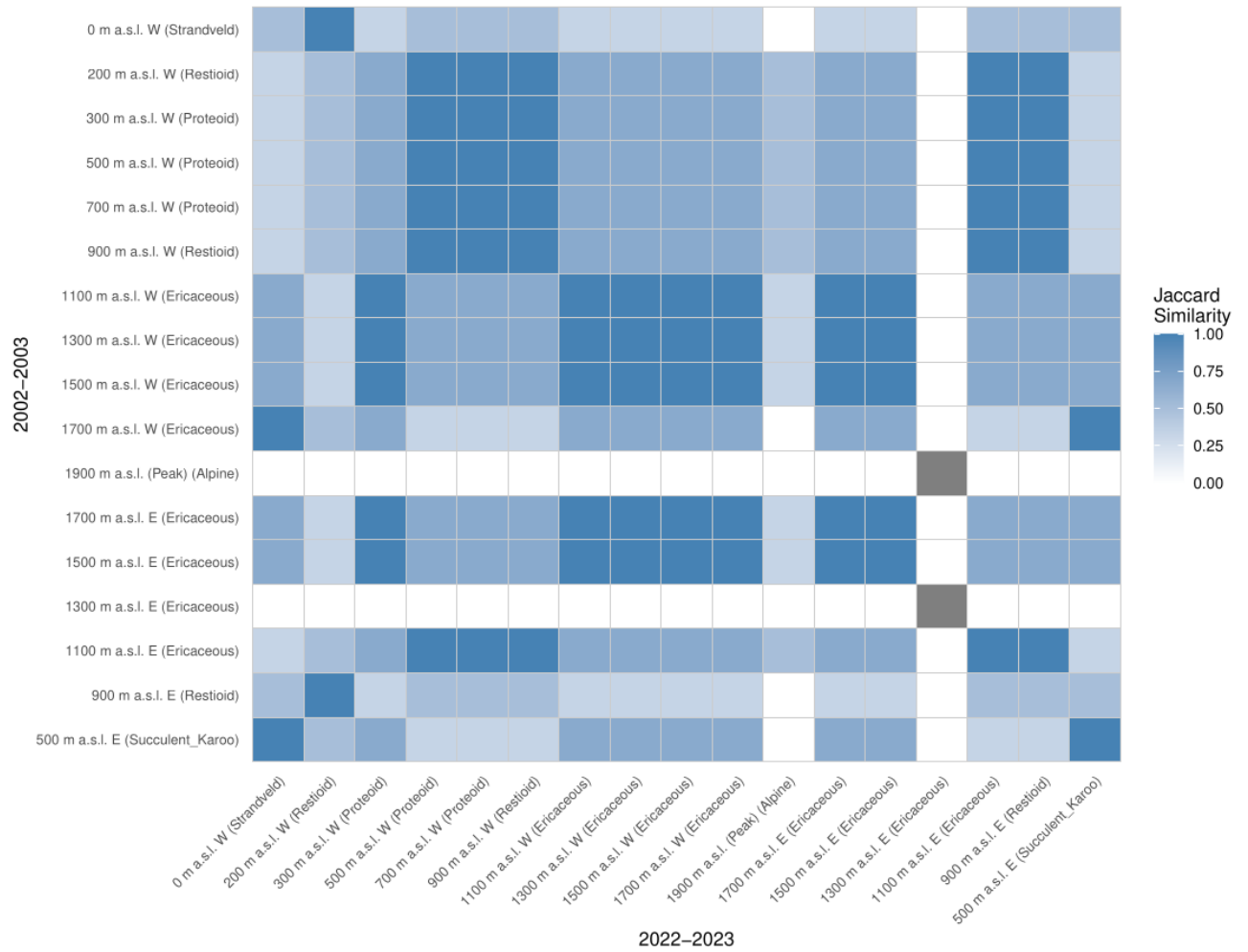


Figure S8: Heatmap of Jaccardian similarities between different altitudinal sites.

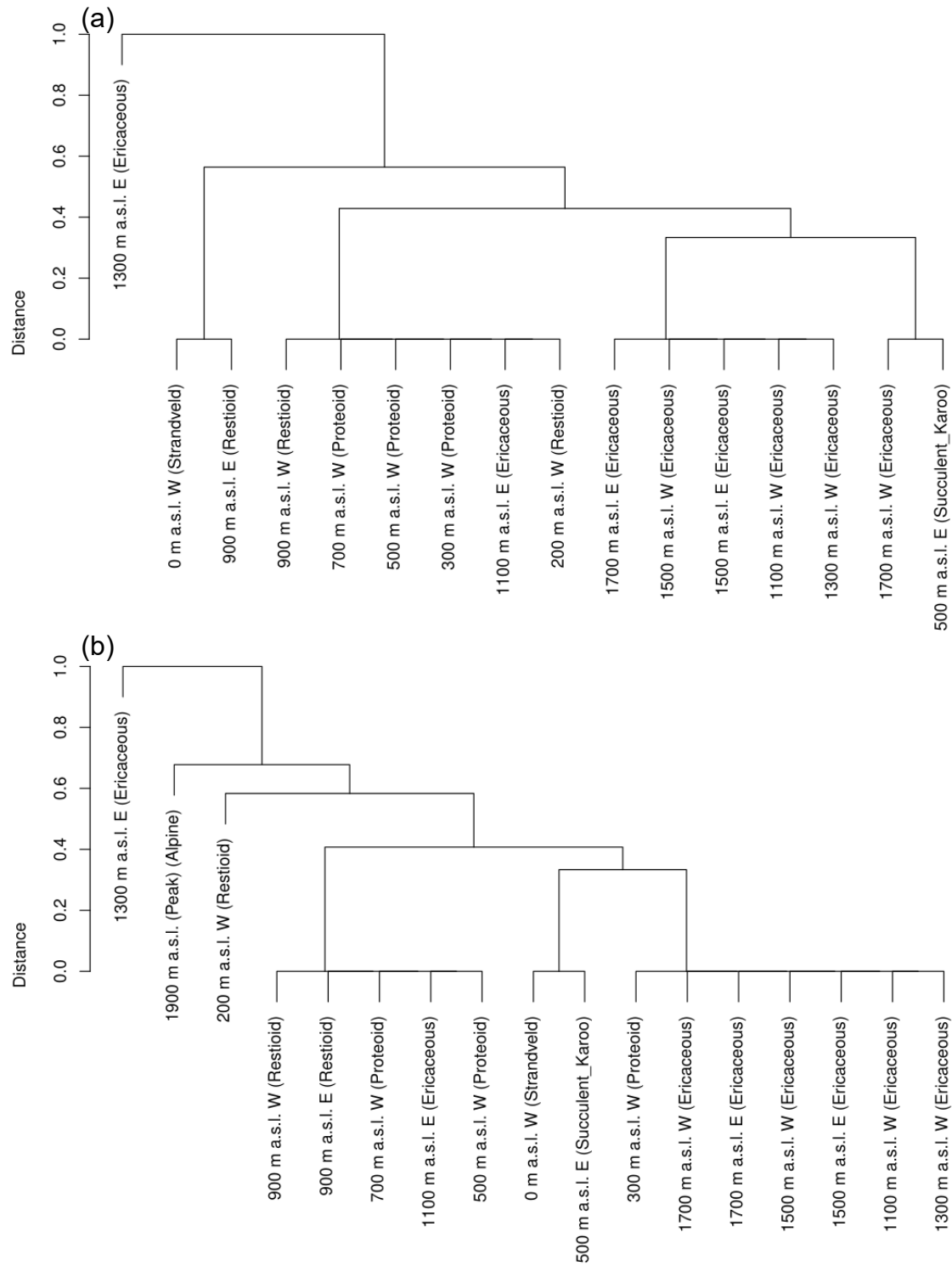


Figure S9: Dendrogram showing the hierarchical clustering of the different sites across the transect in both (a) the 2002-2003 sampling period, and (b) the 2022-2023 sampling period.

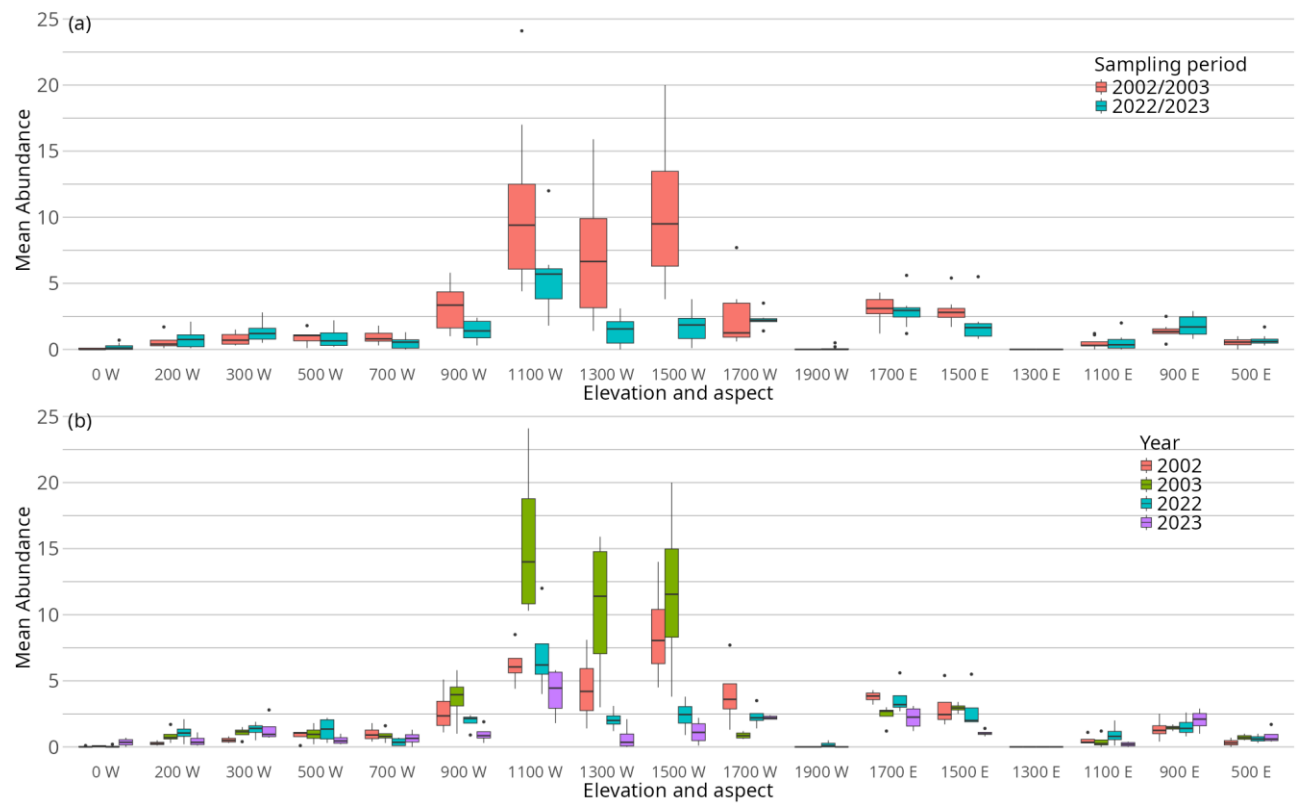


Figure S10: Mean abundance of *Anthia decemguttata*, *Stenocara dentata*, and *Zophosis gracilocornis* separated by (a) sampling period, and (b) year, across altitudinal gradient.